MULTIMODAL COMPUTATIONAL INTELLIGENCE FOR ENHANCING CAREGIVER PERFORMANCE AND CARE DELIVERY

COMMON INTEGRATED SOLUTION REPORT

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DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration. We declare that this common report is our own work and does not incorporate, without acknowledgement, material previously submitted for a degree or diploma at any university or institute of higher learning. To the best of our knowledge, it contains no material previously published or written by another person except where acknowledgement is made in the text. We grant the Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute this report, in whole or in part, in print, electronic, or other media, while retaining the right to use the content in future work.

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ABSTRACT

Residential care homes rely on small teams of caregivers who must keep residents safe, record the care they give, communicate with residents who cannot rely on speech and stay well enough to do all of it. These needs are usually served by separate tools. This report presents CareSense, a web platform that brings four research components into one application for care homes. PULSE estimates caregiver stress risk from wearable data and proposes workload changes for supervisor review. SENTRY detects fall-like movement from camera skeletons. SCRIBE turns spoken care notes into structured records of activities of daily living, with safety alerts. GLOSS tutors caregivers in care-related sign language and in the motor signs of Parkinson's disease.

Each component was built and evaluated on data kept apart from development. The components share one sign-in, role-based access, a caregiver identity, an audit log and a model registry, and they run as modules of one React application served by FastAPI services and Supabase. On unseen data, PULSE reached an ROC AUC of 0.72, SENTRY a sensitivity of 92.3% at a false alarm rate of 1.28%, and SCRIBE lowered word error rate from 26.1% to 15.6%, while GLOSS recognized 59 care signs with a macro-F1 of 0.78. Walkthroughs with care-home staff, caregiver trainees and a sign-language tutor shaped the design of every module. The work shows that one platform can serve four needs, with a person in charge of every decision. Ethics approval and a pilot in a partner care home are the next steps, and a bundle subscription is proposed as the route to adoption.

Keywords: caregiver support, care homes, multimodal AI, decision support, integrated platform

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The project was shaped by people who care for older adults. Mr. Vidyantha Vasana Abeyratne, Chairperson and Founder of Remark Elder Care (Pvt) Limited, and his staff gave us time, insight and a letter of access, and offered to publish a blog article about the research. The Colombo Caregiver & Health Science School hosted our group visit and described how sign language is taught to caregivers. The management, caregivers and residents of the Love and Care Elder Home, including its Director, Mr. Hiran, allowed us several visits between April and August 2026 and shared their records and experience. We also thank Mrs. Buwensawari for helping us understand activities of daily living, and the volunteers who took part in recording sessions.

We acknowledge the owners of the public resources on which the individual components rest: the TILES-2018 data set from the Signal Analysis and Interpretation Laboratory at the University of Southern California, which was released to us under a signed data use agreement; the NTU RGB+D and UR Fall Detection datasets, the former released to us by the ROSE Lab at Nanyang Technological University; the WLASL and ASL Citizen sign-language datasets; and the Svarah, DisfluencySpeech and DISCO speech corpora. The TILES-2018 data collection was supported by the Office of the Director of National Intelligence (ODNI), Intelligence Advanced Research Projects Activity (IARPA), under IARPA Contract No. 2017-17042800005.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADL	Activities of Daily Living
AI	Artificial Intelligence
API	Application Programming Interface
ASL	American Sign Language
AUC	Area Under the Curve
CCHSS	Colombo Caregiver & Health Science School
CEAI	Centre of Excellence for Artificial Intelligence
CI	Confidence Interval
CNN	Convolutional Neural Network
COCO	Common Objects in Context (17-joint skeleton format)
CPU	Central Processing Unit
DTW	Dynamic Time Warping
EMA	Exponential Moving Average
FAR	False Alarm Rate
FR	Functional Requirement
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
LoRA	Low-Rank Adaptation
LSTM	Long Short-Term Memory
NFR	Non-Functional Requirement
NTU	Nanyang Technological University (also the NTU RGB+D dataset)
PDPA	Personal Data Protection Act, No. 9 of 2022 (Sri Lanka)
REST	Representational State Transfer
RGB	Red, Green, Blue
ROC	Receiver Operating Characteristic

Abbreviation	Meaning
SDG	Sustainable Development Goal
SHAP	SHapley Additive exPlanations
SLIIT	Sri Lanka Institute of Information Technology
ST-GCN	Spatial-Temporal Graph Convolutional Network
TCN	Temporal Convolutional Network
TILES	Tracking Individual Performance with Sensors (data set TILES-2018)
UR	University of Rzeszow (UR Fall Detection dataset)
WER	Word Error Rate
WESAD	Wearable Stress and Affect Detection (data set)
WLASL	Word-Level American Sign Language (data set)

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and Context

Residential care for older people rests on a small number of people doing demanding work. During a single shift a caregiver may watch over a dozen residents, remember what each one has eaten and taken, find a way to talk with a resident who can no longer hear, and still be expected to reach a fallen resident within seconds. How well that work is done depends heavily on the caregiver's own condition. Prolonged workload is tied to burnout, and burnout to staff leaving, which is costly for health services [1], [2]. Poor staff wellbeing is also linked with lower patient safety [3], and nurse burnout and job dissatisfaction rise as the number of patients per nurse grows [4].

The demand for this work keeps growing. Populations are ageing in every region of the world, and care systems are expected to adapt [5]. The homes visited by the team in Sri Lanka look after residents whose needs increase with age, and their staff carry that increase in their daily routine.

Four problems come up again and again in this setting. Supervisors rarely learn how a caregiver is coping until that person is absent or resigns. Falls are common among older adults and a major cause of injury [6], [7], and care facilities are among the places where they happen [8]. Documentation is still done largely by hand, and electronic record systems have not removed the time it takes [9]. Hearing loss becomes more frequent with age [10], yet caregivers receive little lasting training in sign language: in one survey of health professionals, 86.1% wished to learn it and only 6.3% had been formally trained [11].

The research project reported here, **Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery** (group R26-DS-010), takes these four problems together. Rather than build four separate tools, the group developed **CareSense**, a single web platform. Caregivers, supervisors and administrators sign in once and reach the support that suits their role.

CareSense has four components, and each was built by one member of the group. **PULSE**, by Dayana Priyadharshani Kumar, looks after the caregiver. It estimates

stress risk from wearable data and shows the factors behind each score, and a supervisor can accept or reject the workload change it proposes. Harishalinee Elangovan's **SENTRY** watches over the resident: cameras in a room are turned into body skeletons on site, and a movement that resembles a fall raises an alert for the responsible caregiver. With **SCRIBE**, developed by Vithursha Prabagaran, a caregiver simply speaks. The note becomes a structured record, safety rules check it for anything urgent, and a handover summary is prepared for the next shift. **GLOSS**, by Kaushalya Nagenthraraja, helps caregivers keep their sign language alive through short webcam practice with explained feedback, while a companion module teaches the motor signs of Parkinson's disease.

The four differ in data and in method, yet they meet in the same place, the supervisor and the caregiver working one shift. Section 2.3 explains how the problem divides among them, and Chapter 3 shows how the platform joins them.

1.2 Stakeholders and User Needs

Table 1.1 lists the people and organizations concerned. The platform has three kinds of user, matching the three roles in the shared application: caregivers, supervisors and administrators. Residents and their families do not use the software, but they are the people it is meant to protect. The remaining stakeholders decide whether a care home can adopt it.

Table 1.1: Stakeholders of CareSense and their main needs

Stakeholder	Type	Main need	Components that serve it
Caregivers	Primary user	Record care quickly; communicate with residents who cannot hear or speak; trust that a fall will be noticed; avoid being judged by a stress score	SCRIBE, GLOSS, SENTRY
Supervisors and charge staff	Primary user	See who is under sustained strain and what could be done; receive timely alerts; read clear handover notes	PULSE, SENTRY, SCRIBE
Administrators	Primary user	Create and manage accounts; control who sees what; keep an audit trail	All (shared platform)
Residents and families	Indirect beneficiary	Safe, continuous and communicative care	SENTRY, SCRIBE, GLOSS
Care home owners and managers	Secondary; likely buyer	Fewer missed events, lower turnover, complete records, low cost	All

Stakeholder	Type	Main need	Components that serve it
Caregiver training schools	Secondary	Practice that continues after the course ends	GLOSS
Regulators and data protection authorities	Secondary	Lawful handling of staff and health data; registered homes	Shared platform

These needs were gathered from visits and discussions with care organizations, from supervisor meetings and from the literature. Table 1.2 records the engagement in date order. The dates of the visits to Remark Elder Care and the Colombo Caregiver & Health Science School come from the signed visit letters (Figure C.1), and the supervisor meetings come from the group's Microsoft Teams records (Figure C.5, Figure C.6, Figure C.7 and Figure C.8).

Table 1.2: Stakeholder engagement during the project

Date	Organization and people	Activity	Informed
27th November and 10th December 2025	Supervisor, Prof. Samantha Thelijjagoda	Follow-up discussions on the topic and on refining the topic assessment form	All
January and February 2026	Mr. Velu Sivakumar, external supervisor	Meetings on daily care work and on which features caregivers would use	All
7th April 2026	Remark Elder Care; the Chairperson and Founder, Mr. Vidyantha Vasana Abeyratne, and staff	Group interview on caregiver strain and a practical response (a substitute caregiver), on residents who communicate through the body, and on early signs of Parkinson's disease	PULSE, GLOSS
22nd April 2026	Colombo Caregiver & Health Science School (CCHSS)	Site visit and discussion with staff and a sign-language tutor on how sign language is taught	GLOSS
April, May, July and August 2026	Love and Care Elder Home; Director Mr. Hiran, management and caregivers	Repeated visits. Handwritten care records were seen with permission, and caregivers explained how they document care and which observations call for an alert. No audio was recorded and no resident was interviewed	SCRIBE, SENTRY
11th May 2026	Supervisor and the three other members	Feedback meeting after the progress presentation	All
August 2026	Remark Elder Care and Love and Care Elder Home	Informal discussion of the intended caregiver-support functions with staff	SENTRY

Date	Organization and people	Activity	Informed
August 2026	A Remark Elder Care representative and an elder-care caregiver	Walkthrough of the PULSE prototype using the 15-person demonstration group, followed by open discussion	PULSE

Three working profiles, shown in Table 1.3, summarize what these conversations revealed about the three user roles. They describe roles and are not portraits of any particular person.

Table 1.3: Working profiles of the three user roles

	Caregiver	Supervisor	Administrator
Typical day	Looks after several residents, writes records by hand, answers calls, hands over at shift change	Oversees a team or unit, assigns work, reads records, decides who covers a gap	Sets up the home on the platform, manages accounts, answers to management
Pain points	Paperwork at the end of a shift; no way to talk to residents who cannot hear or speak; a fall can happen when nobody is in the room	No regular view of how staff are coping; important observations lost at handover	Accounts and permissions kept in several places; no record of who did what
Modules used	Voice Log, Sign & Vitals, alerts on a phone	Deterioration Detection, Fall Detection, Voice Log alerts and handover	Account administration, audit log
Must not see	Other caregivers' stress scores	Staff outside their own team	Not applicable

1.3 Alignment with the Sustainable Development Goals

CareSense is aligned first with **Sustainable Development Goal (SDG) 3, Good Health and Well-Being**, which calls for healthy lives and well-being at all ages. The platform serves the goal from two sides. Caregivers gain support because their wellbeing and workload shape how safe care is, and sustained stress among health workers is linked to poorer outcomes for patients and to staff turnover [3], [4]. Residents gain because caregivers can notice falls sooner, record care more completely and communicate better. The work also fits the aims of the Centre of Excellence for Artificial Intelligence (CEAI) at SLIIT, which promotes the responsible use of artificial intelligence for public benefit. Its design reflects the principles that recur in published AI ethics guidelines, namely transparency, responsibility and privacy [12].

A person stays in control of each consequential decision, every output is explained, and the data of caregivers and residents are kept to a minimum.

Table 1.4 shows what each component adds. The statements describe the contribution each component is designed to make, and Section 7.5 weighs that contribution against the targets of the goals.

Table 1.4: Contribution of each component to the SDGs

Component	Main SDG link	Further links
PULSE	SDG 3: caregiver well-being, and safer care through early notice of sustained strain	SDG 8: more manageable and fairer workloads for caregivers; SDG 9: explainable machine learning applied to a workforce problem
SENTRY	SDG 3: quicker help after a fall	SDG 9: low-cost technology on existing equipment
SCRIBE	SDG 3: complete records, earlier attention to symptoms, clearer handover	SDG 8 (less paperwork), SDG 9 (AI for a paper-based sector), SDG 10 (accent-aware speech recognition)
GLOSS	SDG 3: communication with deaf and non-verbal residents; earlier notice of motor symptoms	SDG 4 (affordable skills practice), SDG 10 (access to care for deaf people)

1.4 Purpose and Scope of the Common Report

This report presents the problem that the four components share and the integrated solution built around it. It explains how the components were joined into one platform, how the platform was evaluated and what the group can claim on that evidence. The detailed research behind each component, which covers its data, models, experiments and component-level tests, is documented separately by the member who owns it. The results of that work appear here only to the extent that the integrated view needs them.

The report covers CareSense as it stood in October 2026: a web application with four modules, a shared sign-in and access model, a shared database and file storage, and the services behind each module. Clinical deployment, medical-device certification and a field trial lie beyond this stage of the work and are set out as future directions in Chapter 8.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall Problem Definition

The overall problem can be stated in one sentence. Caregivers in care homes work under sustained strain while keeping residents safe, keeping accurate records and communicating with residents who cannot rely on speech, yet the support available to them is split across separate tools that share no information and are rarely adapted to care homes. The problem exists whatever technology is chosen. It shows in the daily routines of the homes the team visited, and it would remain even if every tool were digital but disconnected.

2.2 Root Causes and Overall Research Gap

2.2.1 Root causes of the problem

Four root causes were identified, one behind each of the four needs described in Section 1.1. Table 2.1 sets each cause beside the way it is handled in current practice and the component that responds to it.

Table 2.1: Root causes of the overall problem and the component that responds

Root cause	Current practice	Responding component
Strain on caregivers goes unseen	Supervisors rely on observation and on staff saying that they are not coping	PULSE
Falls are noticed late	Bed and chair alarms, wearables, depth or thermal sensors, or nobody watching at night	SENTRY
Documentation is manual	Handwritten files for each resident; typed record systems; ambient scribes designed for doctor consultations	SCRIBE
Communication with non-verbal residents depends on skills that fade	Short sign-language units in courses; general apps without care vocabulary	GLOSS

The review that follows takes the four areas in turn and asks where earlier work stops. Its purpose is to state the contribution of CareSense precisely.

2.2.2 Caregiver strain and workload

Research on stress detection with wearables has produced useful datasets and reviews. WESAD recorded wrist and chest signals in a laboratory and labeled the affect states [13], and a review of the field found that most work still depends on controlled conditions [14]. Heart rate variability is a recognized physiological correlate of stress [15]. Hospital staff are covered by two field datasets, TILES-2018 [16] and a collection of nurses recorded during shifts [17]. Work on healthcare workers has shown that a model trained on one real-world dataset travels poorly to another [18]. Rostering research, for its part, treats the allocation of nurses as a scheduling problem built on rules and preferences [19]. Neither line of work joins the two ends: stress studies stop at a score, and rostering ignores how a caregiver is coping on the day.

2.2.3 Fall detection

Reviews describe a long list of approaches, from wearable accelerometers to depth cameras and ambient sensors, and they agree that cost, comfort and false alarms limit adoption [20], [21], [22]. Early camera work analyzed the deformation of a person's silhouette [23]. Later studies applied convolutional networks to video [24] or paired depth maps with a worn accelerometer on an embedded device [25]. Graph convolutional networks over body skeletons [26] have also been applied to falls [27], and a skeleton removes both the background and the face from the input. Two weaknesses recur. A model trained on one dataset does well on it and poorly elsewhere, which is a general property of dataset bias [28], and frequent false alarms teach staff to ignore alerts [29].

2.2.4 Care documentation

Electronic record systems did not shorten documentation time as hoped [9]. Studies that extract activities of daily living (ADL) [30] from records have worked on text that was already typed [31]. Ambient digital scribes listen to a consultation and draft a note, and they are the closest commercial idea; the early evidence on them concerns clinician workflows [32]. Two cautions apply to care homes. Speech recognizers trained mostly on one accent do worse on others, as work on Indian-accented English shows [33], and general speech models can invent text that nobody spoke [34]. Poor transfer of information at handover is, in addition, a known safety risk [35].

2.2.5 Sign language and communication

Deaf patients meet communication barriers in health care that trained and culturally aware providers can reduce [36], [37]. Large datasets of isolated signs exist [38], [39], and pose-based transformers recognize them well [40]. On the learning side, commercial apps charge a monthly fee for a broad vocabulary, free tools offer practice with AI feedback, and research prototypes serve emergency medical providers [41] or novice signers through games [42]. Studies of feedback for sign learners show that explained corrections help [43], [44]. None of these tools targets the vocabulary of everyday care, and none is connected to a caregiver's workplace record. Distributed practice over time supports retention [45], which gives a tutor placed inside the workplace platform a clear advantage.

2.2.6 Comparison of existing approaches

Table 2.2 sets the main approaches against the needs of a care home. The comparison draws on the sources above and reflects what the published material allows one to verify.

Table 2.2: Comparison of similar work with the care-home need

Area	Representative work	What it offers	What it leaves open for a care home
Strain	WESAD [13]; TILES-2018 [16]; nurse dataset [17]	Labeled data and models for stress from wearables	A score only; poor transfer between datasets [18]; no action for the supervisor
Workload	Nurse rostering [19]	Schedules built from rules and preferences	No reaction to how a caregiver is coping
Falls	Silhouette analysis [23]; CNN on video [24]; depth and accelerometer [25]	Detection on benchmark data or with extra sensors	Cost, privacy of raw video, false alarms, weak transfer between sites
Falls	ST-GCN on skeletons [27]	Detection from body joints only	No fusion with hand-built features; no temporal alert states; no link to care staff
Documentation	ADL extraction from records [31]; ambient scribes [32]	Structured information from text; drafted notes in consultations	Typed input; not designed for care-home activities, accents or safety alerts
Sign learning	Commercial sign apps; EMS tool [41]; PopSign [42]	General or emergency vocabulary with some feedback	No care vocabulary; no Parkinson's education; no supervisor view
Integration	Separate products for each need	Strong single-purpose tools	Four logins, four dashboards and four copies of staff data

Table 2.3 compares capabilities and not papers. It is a qualitative summary of the approaches reviewed, and the entries for CareSense describe the prototype as built.

Table 2.3: Capability comparison of the main approaches

Capability	Stress research	Fall systems	Ambient scribes	Sign apps	CareSense
Estimates caregiver strain with reasons	Partly	No	No	No	Yes
Suggests a reviewable workload change	No	No	No	No	Yes
Detects falls from an ordinary camera without storing video	No	Partly	No	No	Yes
Turns speech into a structured care record with alerts	No	No	Partly	No	Yes
Teaches care-related signs with explained feedback	No	No	No	Partly	Yes
One sign-in and role-based access across all tasks	No	No	No	No	Yes
Keeps a person in control of consequential decisions	Partly	Partly	Partly	Not applicable	Yes

2.2.7 The overall research gap and the novelty of CareSense

Two gaps follow from the review. The first lies inside each area: strain research ends at a score, fall detectors do not carry over between settings, scribes are not built for care homes and sign apps pay no attention to care vocabulary. The second lies between the areas. A caregiver meets all four problems together, but the tools that exist for them are separate. A supervisor therefore cannot see that an overworked caregiver is also the one whose notes are late and whose room has had several movement alerts. A four-component solution is needed because only a joined platform can close both gaps.

Table 2.4 states what CareSense adds. The claim is deliberately limited. Gradient boosting [46], graph convolutions, speech recognition and sequence matching are established techniques. The novelty lies in how they are fitted to the care-home problem and combined behind a shared, role-aware platform that keeps a person in the loop. Among the sources reviewed, none describes a system that joins these four functions for care homes.

Table 2.4: Novel elements of CareSense and their justification

Element	Existing position	What CareSense adds
A common platform for strain, falls, documentation and communication	Separate tools for each area	One sign-in, roles, caregiver identity, audit log and model registry shared by four modules
From stress score to reviewable action (PULSE)	Scores without action; rostering without strain	A two-stage model with SHAP explanations [47], and a rule that proposes a same-unit substitute only after sustained strain
Skeleton fusion with temporal alert states (SENTRY)	Single-model detectors with frame-level alarms	An ST-GCN fused with 18 biomechanical features and smoothed into three alert states, with cross-dataset behavior examined
Care-home voice-to-record pipeline (SCRIBE)	Consultation scribes; typed ADL extraction	A 19-field ADL schema, accent-adapted recognition, a protected vocabulary and field-based alert rules
Care-specific sign tutor with cautious feedback (GLOSS)	General vocabulary; recognition without explanation	59 care signs, a repeat request when unsure, rule-based feedback reworded only behind a guard, and Parkinson's education

2.3 Subproblem and Component Mapping

The overall problem was divided into four subproblems, one for each member. The division is deliberate. Each subproblem rests on a different type of data and a different kind of model, so the components do not overlap, yet all of them depend on the same platform and the same users. Table 2.5 shows the mapping.

Table 2.5: Subproblems and components

Component	Owner	Subproblem	Data type
PULSE	Dayana Priyadharshani Kumar (IT22178640)	Supervisors cannot see sustained caregiver strain or its drivers, nor decide how to respond	Wearable physiological data and daily features
SENTRY	Harishalinee Elangovan (IT22057488)	A fall or risky movement may be noticed late	Camera video, converted to skeletons
SCRIBE	Vithursha Prabagaran (IT22294098)	Care is documented by hand, slowly and inconsistently	Caregiver speech
GLOSS	Kaushalya Nagenthraraja (IT22289384)	Caregivers cannot practice care-related sign language or recognize early motor symptoms	Webcam video, converted to hand and body landmarks

All four are needed. Without PULSE the caregiver's own condition goes unobserved, and without SENTRY the most urgent safety event has no automated response. Documentation stays on paper without SCRIBE, which also leaves the alert rules with nothing to read. Communication with non-verbal residents has no support without GLOSS. The components meet at the supervisor as well: the same person receives the strain view, the fall alerts and the voice-note alerts, so the picture of a shift is built from all of them.

2.4 Overall Aim and Group Objectives

Overall aim: To design, build and evaluate an integrated, privacy-aware platform that helps care-home caregivers and supervisors monitor caregiver strain, detect falls, document care by voice and learn care-related sign language, while keeping a person in control of every consequential decision.

Table 2.6 states seven measurable group objectives. Each is tied to the components that serve it and to the evidence used in Chapter 6 to judge it.

Table 2.6: Group objectives and evidence of achievement

No.	Objective	Components	Evidence of achievement
GO1	Estimate caregiver stress risk from wearable data with an explainable model and propose reviewable workload changes	PULSE	Held-out ROC AUC with interval; explanation of each score; supervisor review of suggestions
GO2	Detect fall-like movement from camera skeletons quickly and with few false alarms, without keeping raw video	SENTRY	Sensitivity, false alarm rate and AUC on unseen subjects; latency; privacy design
GO3	Turn spoken care logs into structured ADL records and safety alerts	SCRIBE	Word error rate, extraction accuracy, alert recall on test and safety sets
GO4	Teach care-related signs and the motor signs of Parkinson's disease with explained, cautious feedback	GLOSS	Macro-F1 on unseen signers; calibration; feedback safety checks
GO5	Integrate the four modules into one application with a single sign-in, role-based access, a shared caregiver identity, an audit log and a model registry	All	Running integrated application; integration checks (Section 6.2)
GO6	Keep people in control and protect data: advisory outputs, deferral when uncertain, minimal data, access control	All	Design evidence; security checks; data protection review

No.	Objective	Components	Evidence of achievement
GO7	Evaluate on held-out data, report results with their limits and set out a route to validation and adoption	All	Held-out testing, stated limits, ethics status, commercial analysis (Chapter 7)

2.5 Scope, Assumptions and Constraints

The points below apply to the whole platform and mark the boundaries within which every result in this report should be read.

- **Setting.** Small and medium care homes in Sri Lanka, using a shared web application on a laptop or a phone browser.
- **Advisory role.** Every module supports a decision and does not take one. PULSE assigns no staff, SENTRY does not call for help by itself, SCRIBE records are open to review, and GLOSS teaches without diagnosing; its Parkinson's content is educational.
- **Data.** The models were trained on public or de-identified data: TILES-2018 for PULSE, NTU RGB+D and UR Fall for SENTRY, public speech corpora and a de-identified ADL set for SCRIBE, and WLASL and ASL Citizen for GLOSS. The platform itself runs on synthetic or de-identified data until approval is granted for collecting data in care homes.
- **Ethics and approval.** The ethics application is under review. Until it is decided, no data about identifiable care recipients or caregivers is collected.
- **Infrastructure.** Services run on Render and the front end on Vercel, with Supabase and Cloudflare R2 for data and files. The cost and availability of these services shape the size of any pilot.
- **Exclusions.** The platform is not a medical device and makes no clinical claim. A field trial, clinical validation and a paid release are separate stages that follow this work.
- **Dependencies.** The shared parts (sign-in, caregiver identity, audit log, model registry and storage) must work for any module to be usable. SENTRY also needs a camera and a computer on site.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated Solution Overview

CareSense is one web application for care homes. It is not four products sitting side by side. A user signs in once, and the role stored in the shared profile (caregiver, supervisor or administrator) decides which modules appear in the sidebar and what each module may show. Behind the screens, every module has its own service and models. All of them lean on a common set of services: authentication, a caregiver identity, an audit log, a registry of models, the database and file storage.

The value of the platform lies in what it does for one caregiver's day. A supervisor sees who is under sustained strain and receives a suggestion for sharing the load (PULSE). The same supervisor is alerted when a resident may have fallen (SENTRY) or when a spoken note describes something that needs attention (SCRIBE). The caregiver speaks a record instead of writing it (SCRIBE) and practices, between tasks, the signs needed to talk with residents who cannot hear or speak (GLOSS). The outcome is a care team that knows more about itself, its residents and its records, with a person making every decision.

3.1.1 The shared web application

Every user begins at the sign-in page (Figure 3.1). After sign-in the application reads the user's role and builds the sidebar. The Sign & Vitals entry opens a short chooser that separates sign-language practice from the Parkinson's education module (Figure 3.2). The screenshots in this report were taken from the integrated application in October 2026 using demonstration data and, unless a caption says otherwise, an administrator account. Personal details on the sign-in page have been masked.

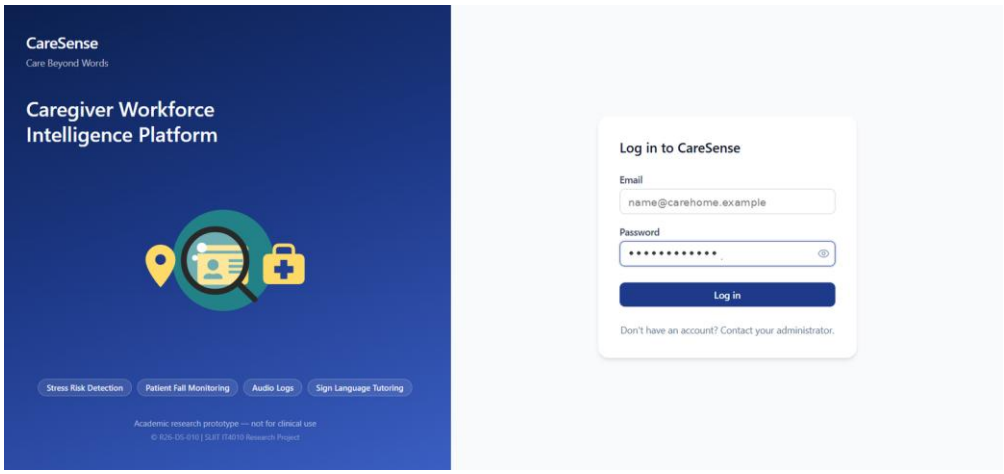


Figure 3.1: Sign-in page of the integrated application (e-mail and password masked)

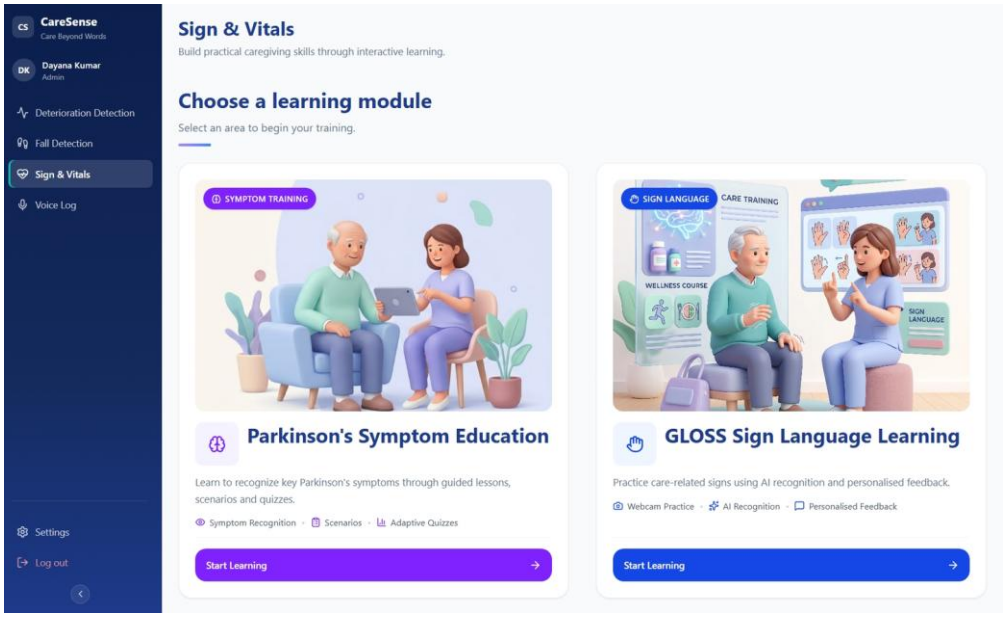


Figure 3.2: Sign & Vitals chooser showing the two learning modules

3.1.2 Functional scope

The platform is specified by fifteen functional requirements, which Table 3.1 lists. The first four apply to every user and every module. The other eleven describe what each module does. Section 6.2 links every requirement to the test that supports it.

Table 3.1: Functional requirements of the integrated platform

ID	Requirement	Module
FR-01	A user signs in with an e-mail address and password and receives a role (caregiver, supervisor or administrator)	Application

ID	Requirement	Module
FR-02	The sidebar and every API call expose only what the role permits; a supervisor sees only their own team	Application, all
FR-03	The same caregiver identity is used across the four modules	Shared
FR-04	Significant events are written to the audit log, and each model in use is listed in the model registry	Shared
FR-05	PULSE scores a caregiver's day and shows a risk tier with the main contributing factors	PULSE
FR-06	PULSE flags sustained risk (at least 3 of the last 5 assessments elevated), proposes a same-unit candidate below 0.35 and records the supervisor's review	PULSE
FR-07	SENTRY analyzes each camera stream and shows a NORMAL, MODERATE or HIGH state live	SENTRY
FR-08	SENTRY stores alerts, lets staff acknowledge them and sends them to the assigned caregiver's phone	SENTRY
FR-09	SENTRY lets staff replay the skeleton of an event for review	SENTRY
FR-10	SCRIBE records a voice note, transcribes it, cleans it and extracts a 19-field ADL record	SCRIBE
FR-11	SCRIBE applies alert rules to the record and produces a written and spoken handover	SCRIBE
FR-12	GLOSS recognizes a sign from a webcam attempt and asks for a repeat when confidence is below 0.35	GLOSS
FR-13	GLOSS gives explained execution feedback, recommends the next lesson and keeps progress and history	GLOSS
FR-14	GLOSS teaches six Parkinson's motor symptoms with a quiz and makes no diagnostic claim	GLOSS
FR-15	Administrators assign roles and caregiver room assignments	Application, SENTRY

3.1.3 Application of data science principles

CareSense is a data science product from end to end, and each component follows the stages of a data science workflow. Table 3.2 shows how the main pillars were applied. The same stages were kept with the same discipline in all four components, which is what allows the results in Chapter 6 to be read side by side.

Table 3.2: Data science pillars applied in the four components

Pillar	PULSE	SENTRY	SCRIBE	GLOSS
Data acquisition and governance	TILES-2018 [16] under a signed data use agreement; 15 people kept for demonstration	NTU RGB+D [48] released on request; UR Fall; webcam clips from two volunteers	Public speech corpora; de-identified ADL text; 200-case safety set	WLASL [38] and ASL Citizen [39]; one landmark pipeline for all videos

Pillar	PULSE	SENTRY	SCRIBE	GLOSS
Preparation and features	Daily wearable features; expected heart rate and its residual; 16 Stage-2 inputs	14-joint skeletons in 90-frame windows; 18 biomechanical features	30-second chunks; sentence splitting; 19-field ADL schema	59 care signs kept from 66; landmarks normalized
Modeling	Two-stage gradient-boosted trees [46]	ST-GCN [26] with late fusion	Whisper-small [49] with LoRA [50]; BART-base [51]; T5-small [52]	TCN [53] and SPOTER-style transformer [40] ensemble
Evaluation	Split of 140, 30, 27 and 15 people fixed before features; grouped cross-validation; bootstrap interval	Subject-disjoint split; cross-dataset test; confidence intervals	Held-out test and safety sets; word error rate; exact match; alert recall	Unseen signers; macro-F1 with interval; calibration
Explainability and trust	SHAP factors [47]; wording that is not a diagnosis	Three alert states; skeleton replay	Alerts tied to named fields; records open to review	Rule-based feedback; guard on rewording; repeat when unsure
Deployment and traceability	Model checked by hash, feature order and frozen predictions	Saved model files with fixed post-processing thresholds	Model registry entry	Model registry entry; versioned migrations
Ethics and privacy	Display identities synthetic; staff-only view	No raw video stored; faces blurred	De-identified training text	No video stored

3.2 High Level Architecture

Figure 3.3 shows the architecture in layers. The browser runs a React and TypeScript application, hosted on Vercel, which holds the shared sign-in and the four module screens. Each module calls its own FastAPI service on Render. The services rely on Supabase for authentication and for the PostgreSQL database with row-level security, and Cloudflare R2 keeps the files that are too large for the database: SENTRY replay frames, SCRIBE audio and GLOSS reference videos. Beyond the hosted platform, SENTRY depends on equipment in the care home, namely a camera and a computer that extracts skeletons, and it sends alerts to a mobile client through a WebSocket. Two optional external services complete the picture. A language model rewords GLOSS feedback, and a text-to-speech service voices SCRIBE's handover.

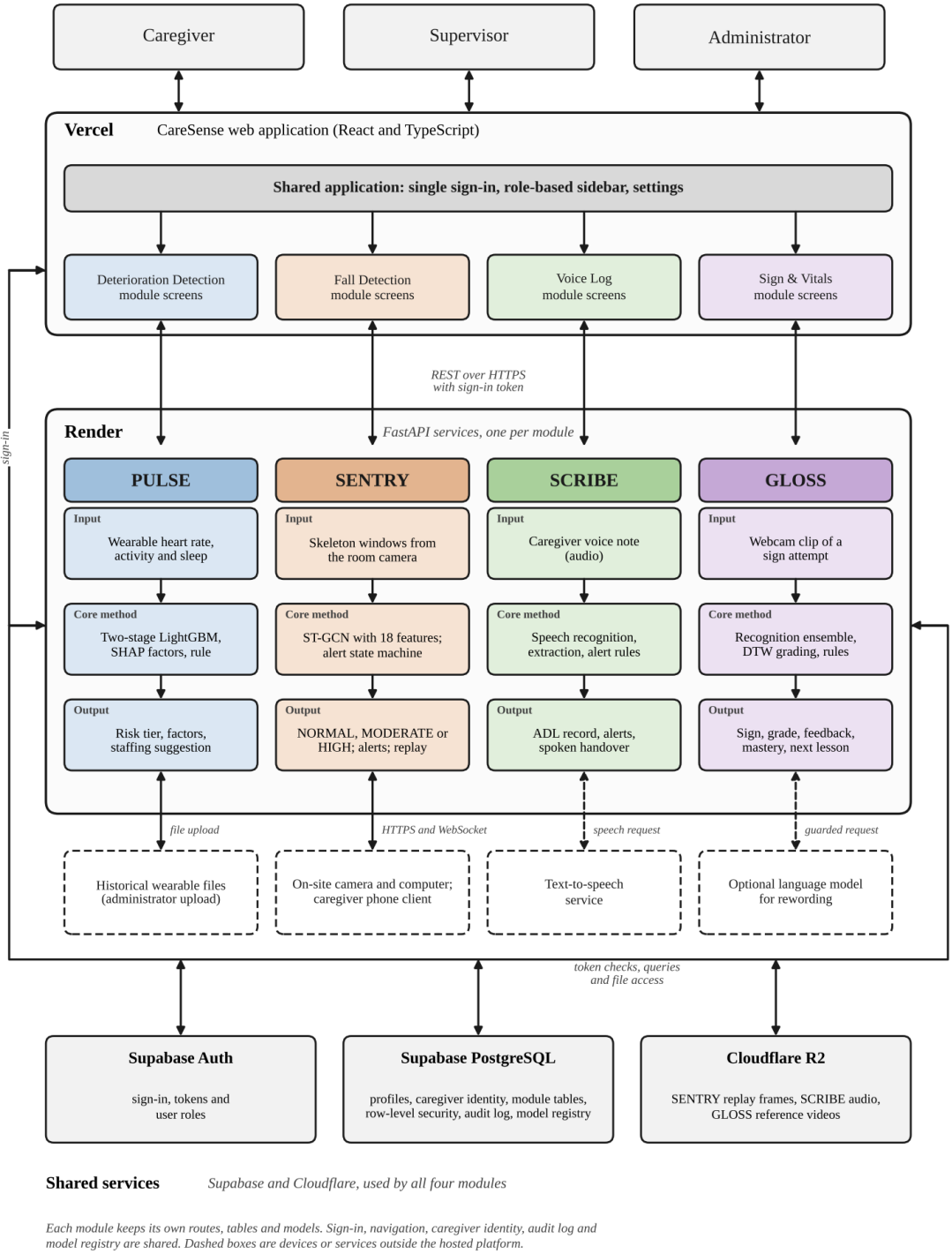


Figure 3.3: High-level architecture of CareSense

The boundaries follow a rule that was set early: each module owns its tables, routes and models, and anything shared is defined once. Table 3.3 lists what is shared and

what stays private. The rule kept the modules independent, and the integration confirmed it, since the PULSE routes, tables and models needed no change when the other components were added.

Table 3.3: Shared and module-specific parts of the platform

Shared by all modules	Private to a module
Single sign-in (Supabase Auth) and role in the profiles table	Models, thresholds and training artifacts
Role-based sidebar and navigation	FastAPI routes and services
caregiver_profiles: the caregiver identity linked to PULSE, GLOSS, SENTRY and SCRIBE	Module tables (for example fall events, gloss_ and pdedu_ tables)
audit_log: one append-only event record	Module-specific object storage paths
model_registry: component, model name, version, metrics	Module-specific alert and feedback rules
Supabase database and Cloudflare R2 storage	Module-specific tests

3.2.1 External interfaces and communication

The platform talks to a small number of external systems. Table 3.4 states what passes across each boundary and how it is protected. The camera path of SENTRY deserves a note. The data that leave the room are skeletons and not video, so the privacy argument rests on where the skeleton extraction takes place. Processing at the edge in this way keeps raw data close to its source [54].

Table 3.4: External interfaces of the platform

Interface	Protocol and direction	Data exchanged	Protection
Users to the front end (Vercel)	HTTPS, browser to host	Pages and scripts	No secrets in the browser bundle
Front end to module services (Render)	REST and JSON over HTTPS; bearer token on every call	Requests and results of the four modules	Role checked by each service; cross-origin access limited to the hosted front end
Services to Supabase	Client library over HTTPS	Tokens, profiles, module tables, audit and registry rows	Row-level security; service-role key kept in host environment variables
Services to Cloudflare R2	Object storage API	SENTRY replay frames, SCRIBE audio, GLOSS reference videos	Module-specific paths; access only from the services

Interface	Protocol and direction	Data exchanged	Protection
Administrator to PULSE (data upload)	REST over HTTPS; file upload by an authenticated administrator	Heart-rate, activity-summary and survey files for monitored caregivers	Administrator role only (any other role gets a 403); a file missing a required column is rejected with a 400; staged data are revealed one day at a time
Room camera and on-site computer to SENTRY	Video stays local; skeleton windows are sent over an authenticated HTTPS and WebSocket connection	14-joint skeletons in 90-frame windows	Raw video not stored; faces blurred in the live stream
SENTRY to the mobile client	WebSocket /ws/alerts; REST	Alert messages for the assigned caregiver	Authentication required in the integrated application
GLOSS to a language model (optional)	HTTPS API call	Rule-written feedback text to be reworded	Called only when confidence is high enough; output checked by a guard; the module works without it
SCRIBE to a text-to-speech service	HTTPS call to a web service	Handover text	Free service in the prototype; a licensed service is planned for commercial use

3.3 Component 1 Summary: PULSE

PULSE is the work of Dayana Priyadharshani Kumar. Table 3.5 gives its summary in the form used for all four components, so that they can be compared directly.

Table 3.5: Summary of PULSE

Item	Summary
Owner	Dayana Priyadharshani Kumar (IT22178640)
Subproblem	Supervisors cannot see which caregivers are under sustained stress, what lies behind it, or how to respond.
Input	Daily features from wearable data (heart rate and related measures, activity, sleep), the previous self-reported stress response, and baseline values for each person.
Core method	A two-stage gradient-boosted model [46]. Stage 1 learns the expected heart rate; its residual and 15 other inputs then give the probability of elevated stress in Stage 2. SHAP values [47] explain each score. A rule flags a caregiver when at least 3 of the last 5 scored assessments are elevated, and then searches the same unit for a candidate whose score is below 0.35.
Output	Risk tier (high at 0.5 or above, moderate from 0.35, low below 0.35), the main contributing factors, trends and a reviewable redistribution suggestion.
Users	Supervisors and administrators only; caregivers cannot see these scores.

Item	Summary
Contribution to the platform	The caregiver-wellbeing view, and the route from a stress score to a staffing suggestion that a supervisor can accept or reject.
Sidebar name	Deterioration Detection

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Figure 3.4 shows the overview that a supervisor sees first. Cards count the caregivers in each risk tier, and an attention list names those who need a conversation, each with the latest score and its change. The Risk Analysis tab (Figure 3.5) opens one caregiver's assessment. It separates the factors that raise the score from those that lower it, plots the risk history, and states that the factors describe the model's prediction and not a diagnosis. The caregiver names are synthetic display identities.

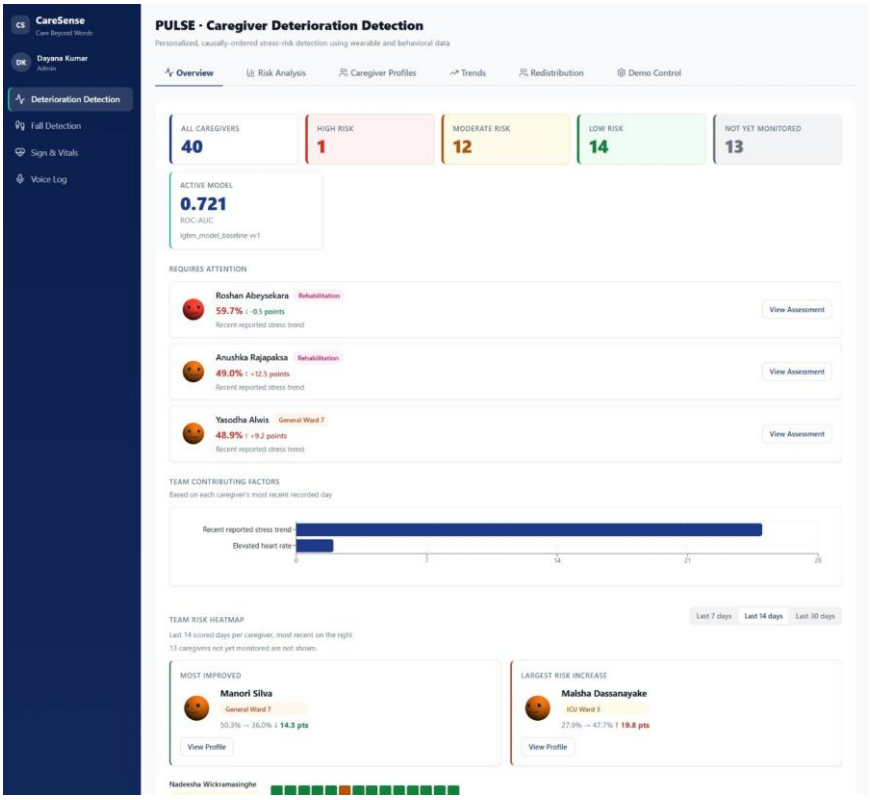


Figure 3.4: PULSE overview: caregivers by risk tier, attention list and team factors

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

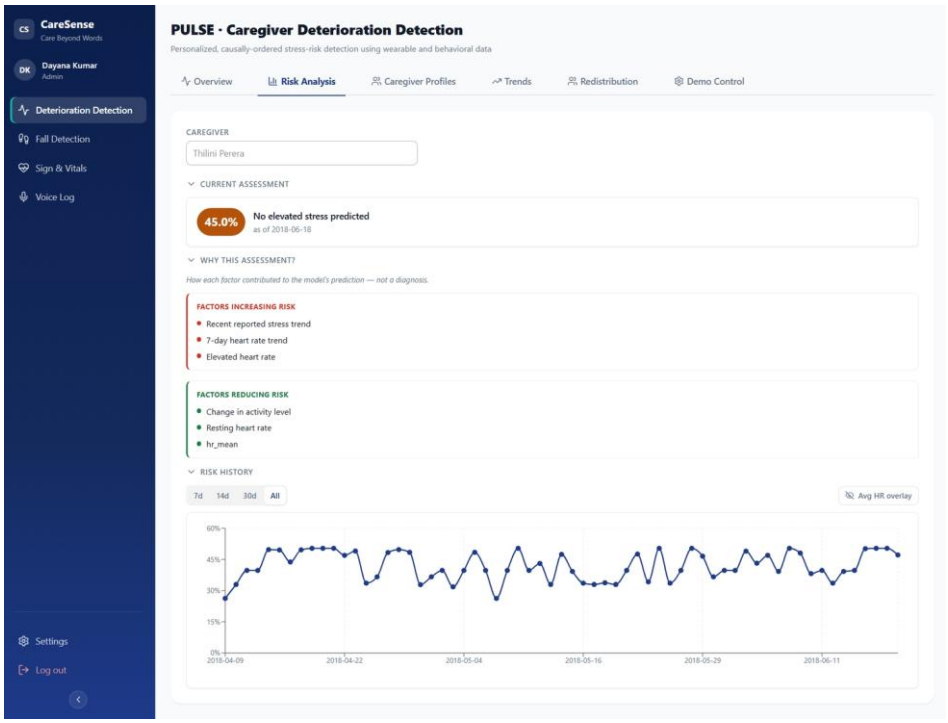


Figure 3.5: PULSE risk analysis of one caregiver, with the factors that raise and lower the score

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

3.4 Component 2 Summary: SENTRY

SENTRY is the work of Harishalinee Elangovan and follows the same structure (Table 3.6).

Table 3.6: Summary of SENTRY

Item	Summary
Owner	Harishalinee Elangovan (IT22057488)
Subproblem	A fall or risky movement in a room may be noticed late, and existing sensors involve trade-offs in cost, comfort or false alarms.
Input	Camera frames, converted on site to 14-joint skeletons in windows of 90 frames (3 seconds). Raw video is not stored, and faces are blurred in the live stream.
Core method	Late fusion of an ST-GCN embedding [26] with 18 biomechanical features, followed by temporal post-processing that turns window scores into NORMAL, MODERATE and HIGH states.
Output	Live state per room on a dashboard through WebSocket, stored fall events and alerts, skeleton replay for review, and alerts sent to a mobile client.
Users	Supervisors and ward managers who review incidents; caregivers who receive alerts.
Contribution to the platform	The safety view: the module that turns a camera in a room into an alert for the responsible caregiver.

Item	Summary
Sidebar name	Fall Detection

The dashboard (Figure 3.6) summarizes alerts by risk level, the number of rooms with a live camera and the average risk score over the last twelve hours. Live Monitoring (Figure 3.7) draws the skeleton of the person in the selected room and lists the state of the other rooms beside it, so that a HIGH alert can be acknowledged from one place. Alert counts on both screens come from demonstration data.

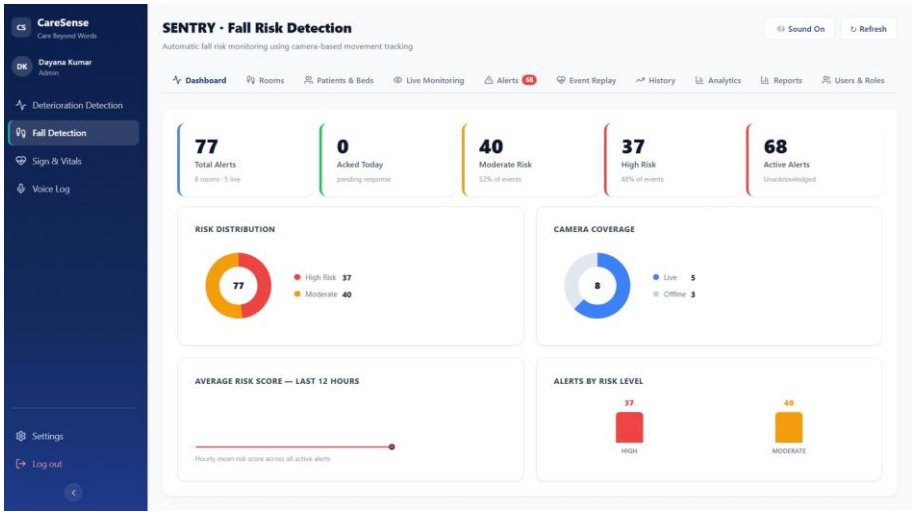


Figure 3.6: SENTRY dashboard with alert counts, room coverage and risk distribution

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

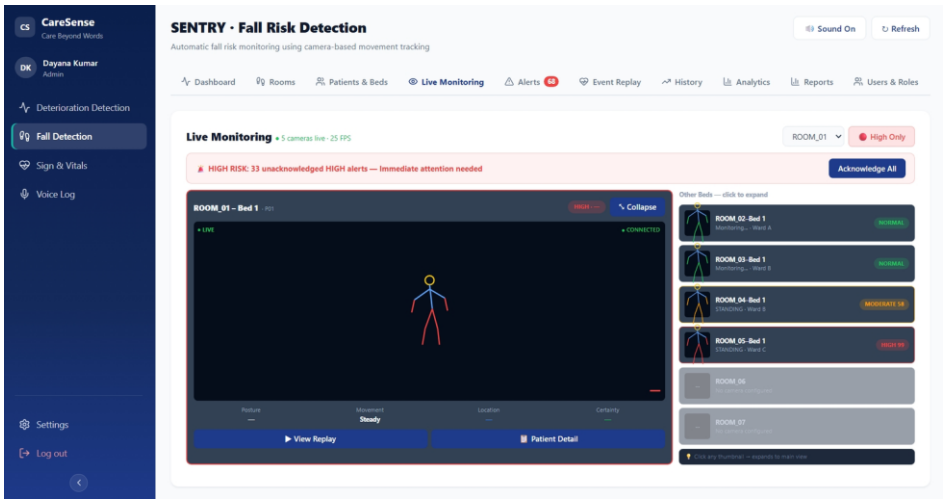


Figure 3.7: SENTRY live monitoring with the skeleton view and the state of each room

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

3.5 Component 3 Summary: SCRIBE

SCRIBE is the work of Vithursha Prabagaran. Table 3.7 gives its summary.

Table 3.7: Summary of SCRIBE

Item	Summary
Owner	Vithursha Prabagaran (IT22294098)
Subproblem	Care is documented by hand, slowly and inconsistently, and observations are missed or not passed on at handover.
Input	A caregiver's spoken note, recorded in the browser and cut into chunks of up to 30 seconds.
Core method	An eight-stage pipeline: chunking; Whisper-small [49] with LoRA [50] for speech recognition; sentence splitting; BART-base [51] for disfluency removal; grammar correction with a protected vocabulary; T5-small [52] extraction of a 19-field ADL record; field-based alert rules; and templated written and spoken handover.
Output	A structured ADL record, an alert decision, and a written and spoken handover report, stored in the shared database (audio in R2).
Users	Caregivers who speak; supervisors who receive alerts and handovers.
Contribution to the platform	The documentation layer: care records and alerts that supervisors read in the same place as the strain and fall information.
Sidebar name	Voice Log

In Voice Log the caregiver selects a resident and starts a care activity (Figure 3.8), then narrates only the care details, such as medication, meals and hygiene. The History tab (Figure 3.9) groups the recordings by date and resident, shows the transcript and category of each observation, and marks the residents whose notes contain an entry flagged for the supervisor. Residents appear under codes such as P01, and the entries are demonstration data.

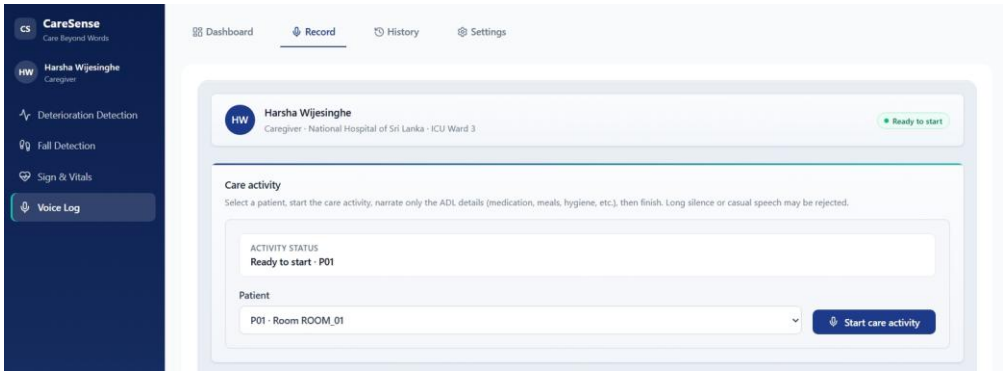


Figure 3.8: SCRIBE recording tab

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

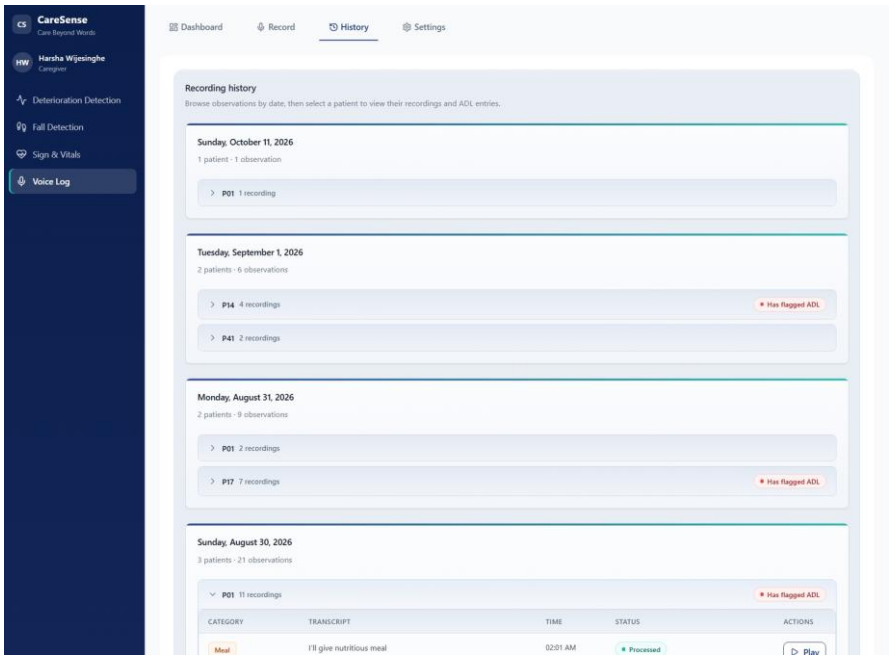


Figure 3.9: SCRIBE recording history with transcripts and flags

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

3.6 Component 4 Summary: GLOSS

GLOSS is the work of Kaushalya Nagenthraraja. Table 3.8 gives its summary.

Table 3.8: Summary of GLOSS

Item	Summary
Owner	Kaushalya Nagenthraraja (IT22289384)
Subproblem	Caregivers forget the sign language taught on courses, and early motor signs of Parkinson's disease are easily missed.
Input	A webcam recording of a practice attempt, converted on the server to hand and body landmarks (no video is stored).
Core method	An ensemble of a temporal convolutional network [53] and a SPOTER-style transformer [40] recognizes 59 care signs. Dynamic time warping [55] against a reference sign gives execution feedback. Rules generate the correction text, and an optional language model rewords it behind a guard [56]. Predictions below 0.35 confidence lead to a request to repeat. A separate education module covers six motor symptoms of Parkinson's disease [57] and makes no diagnostic claim.
Output	Recognition result, explained feedback, next-lesson recommendation, progress record and quiz results.
Users	Caregivers and trainees; supervisors can view progress.
Contribution to the platform	The communication layer: skills practice that continues after the course, inside the system that caregivers already use.
Sidebar name	Sign & Vitals

The practice home (Figure 3.10) recommends the next sign from the caregiver's recent practice and shows mastery, the number of signs practiced and a practice streak. After an attempt, the result screen (Figure 3.11) reports what the model recognized, records the mastery status and offers a repeat or the next recommended sign. The Parkinson's module has its own screens, which Appendix E shows.

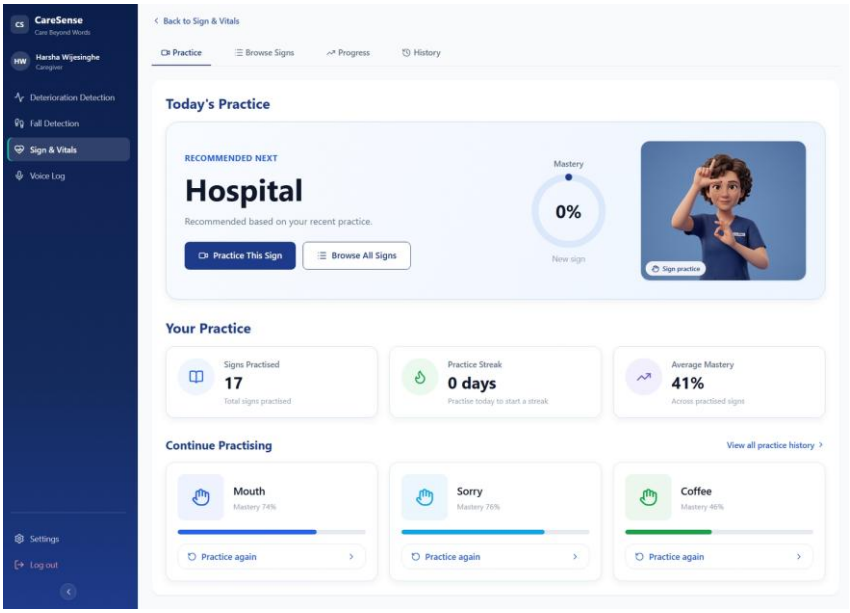


Figure 3.10: GLOSS practice home with the recommended sign

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

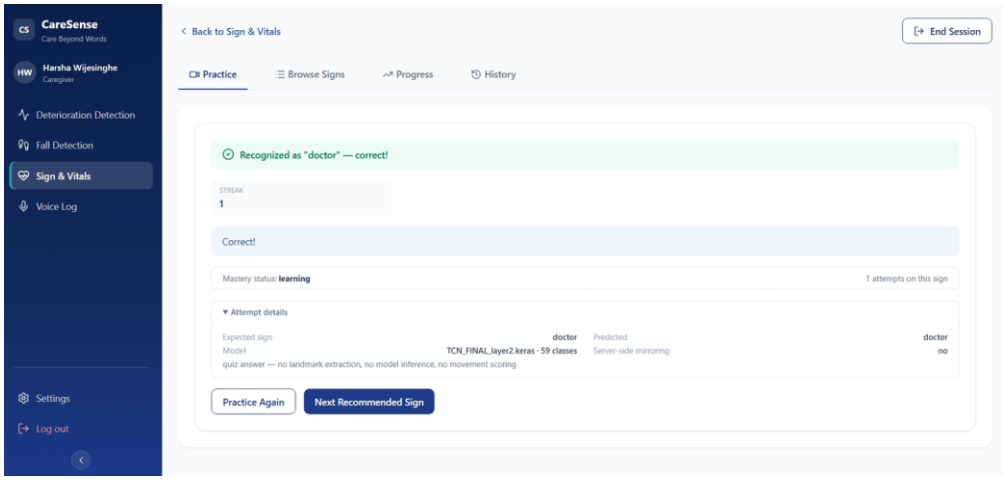


Figure 3.11: GLOSS result after a practice attempt, with recognition details

Source: Screenshot of the CareSense application, Oct. 2026. Names and records shown are demonstration data.

3.7 End to End User and Data Flow

Figure 3.12 follows each kind of data from its source to the person who sees it. The four paths differ in what they compute, but they begin and end in the same way, with one sign-in, one caregiver identity and one audit trail.

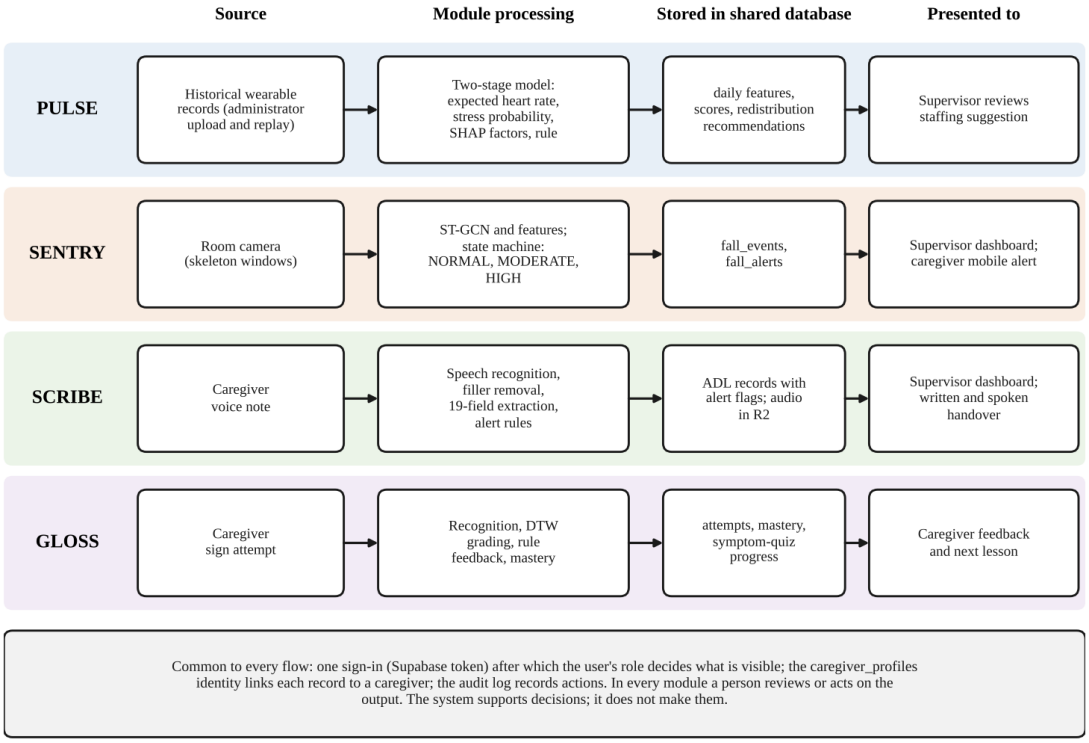


Figure 3.12: End-to-end data flow through the four modules

A typical shift shows how the parts work together. A caregiver signs in once and opens Voice Log to speak a note about a resident's meal and medication. SCRIBE turns it into a record, decides whether anything needs attention and raises an alert for the supervisor if so. Overnight, the camera in a resident's room feeds skeletons to SENTRY. When the movement looks like a fall, the state moves to HIGH after about 2.5 seconds, the dashboard shows it and the assigned caregiver is alerted on a phone. In the morning the supervisor opens Deterioration Detection, sees which caregivers have shown sustained strain, reads the main factors and decides whether to act on the suggested cover. Between tasks the caregiver opens Sign & Vitals, practices a sign in front of the webcam and receives feedback that explains what to change. Every step is written to the audit log.

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration Strategy

Integration followed an interface-first approach in five steps.

- **Components first.** Each member built and tested a stand-alone component with its own FastAPI service, tables and models, so that the four lines of research could proceed in parallel.
- **Shared contracts.** The team agreed on what every module would depend on: Supabase Auth with a role in the profiles table, the caregiver_profiles identity, one audit_log definition and one model_registry. Module tables carry their own prefixes, for example gloss_ and pdedu_, so that no module can overwrite another.
- **Shared web application.** The React application, with the sign-in, the role-based sidebar and the navigation, was built by dividing the work by part among the four members.
- **Module registration.** Each module joined the application as a route group that calls its own service, so adding one module needed no change to the others. The sidebar names are Deterioration Detection, Fall Detection, Voice Log and Sign & Vitals.
- **Hosted environment.** The front end is hosted on Vercel and the services on Render, with Supabase and R2 as shared back-end services. All work is held in one repository, R26-DS-010-Intelligent-Care-Support (Section 5.2).

Four risks were managed along the way.

- The first was that a fault in one module could spread to the others. Module boundaries were therefore kept narrow: PULSE, SENTRY, SCRIBE and GLOSS each own their routes, tables and models, and none calls another module's service.
- The second was security. Before the integrated release, every module was brought under the same sign-in and role checks. The stand-alone SENTRY prototype had no authentication on its API and WebSocket, and this was secured in the integrated application. The PULSE and GLOSS services already used authentication and role checks. SCRIBE holds care records and caregiver audio, which are sensitive. It is

therefore reachable only through the shared sign-in, and its tables and audio files stay under its own paths.

- The third was that a score or record could not be traced to its source. Every model behind a result is recorded in the registry, so a PULSE score, a SENTRY alert, a SCRIBE record or a GLOSS judgement can be tied to the model that produced it.
- The fourth concerned SCRIBE. A speech and language model can mishear a word or leave out an observation, and a care record is relied on at handover. Two safeguards limit this. Field-based alert rules act as a backstop and raise an alert even when the extractor misses a symptom, and the written and spoken handover is generated from templates, so it cannot state anything that is not in the stored record. Records also remain open to review by a person.

4.2 Interfaces, Data Contracts and Shared Models

Table 4.1 lists the interfaces between the front end, the modules and the shared services, and Figure 4.1 shows the shared data model. Appendix A gives the interface specification in more detail.

Table 4.1: Interfaces between the front end, modules and shared services

Interface	Type	Contract	Owner
Front end to module services	REST over HTTPS	Bearer token issued by Supabase Auth on every request; each service checks the role from the profile	Each module
SENTRY live results	WebSocket /ws/live/{room_id}	Live state, scores and skeleton for one room	SENTRY
SENTRY alerts	WebSocket /ws/alerts	Alert messages used by the dashboard and the mobile client	SENTRY
Caregiver identity	Table caregiver_profiles	One row per caregiver, used by all four modules; SENTRY reads it as an external table, read-only	Shared
Audit trail	Table audit_log	Append-only event record shared by all modules	Shared
Model traceability	Table model_registry	Component, model name, version, training time, metrics	Shared
Files	Cloudflare R2	SENTRY replay frames, SCRIBE audio, GLOSS reference videos, with module-specific paths	Each module

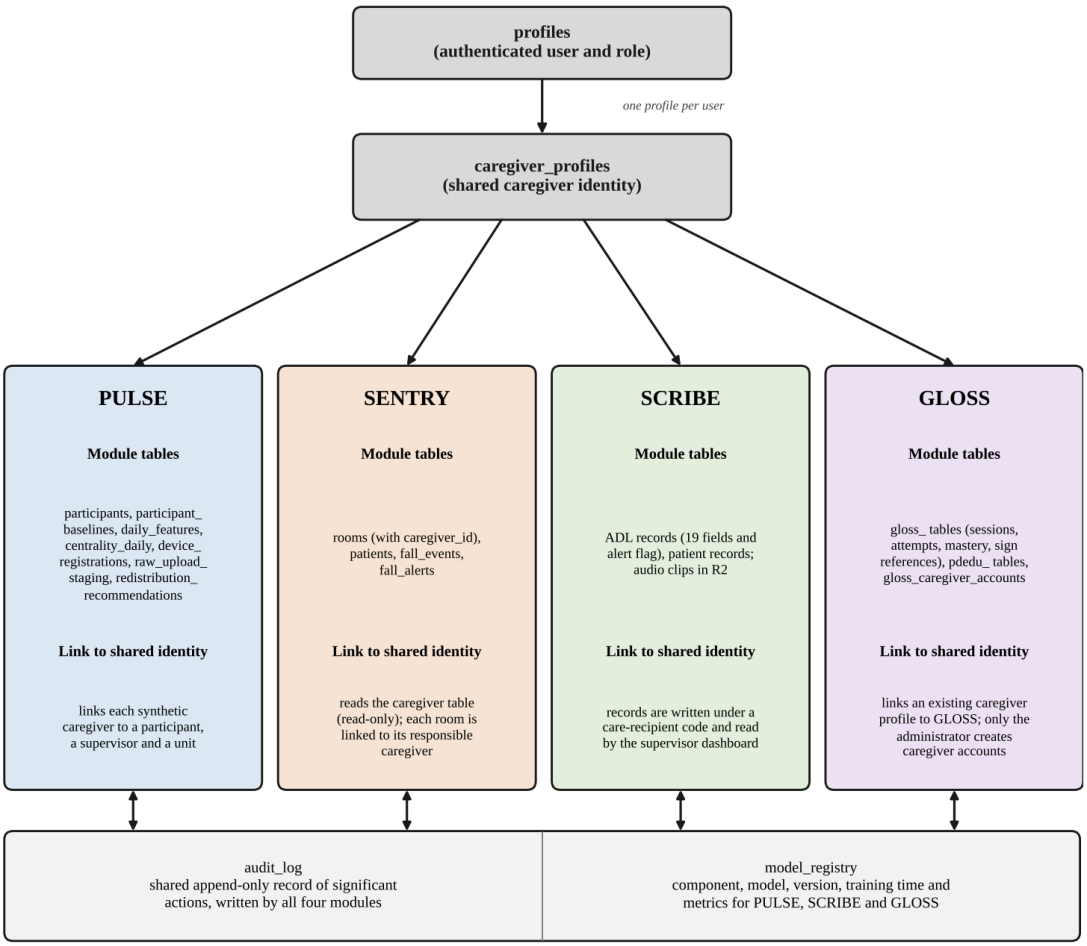


Figure 4.1: Shared data model and module tables

The services expose different numbers of routes. PULSE has 18 routes in six groups: service, caregivers, scoring, analytics, redistribution and administration. SENTRY has routes for rooms, patients, alerts, replay, caregivers and offline analysis, besides the live video stream and its two WebSockets. GLOSS offers routes for practice attempts, next-lesson recommendation, reference animations and the Parkinson's quiz. SCRIBE's routes follow the functions seen in its interface: recording, processing, history, dashboard summary and handover brief.

All services return errors in the format defined for PULSE, a JSON body with a message and a standard HTTP status code. A status of 400 means that the input was invalid, 403 that the role does not permit the action, 404 that the item does not exist,

and 500 that an unexpected failure occurred. Each service validates its own input, so the front end passes on only what the user entered or recorded.

4.3 Shared Nonfunctional Requirements

The requirements in this section apply to every module. They were written for the platform as a whole and sit above the requirement lists of the individual components, and each one is tied to a verification in Section 6.2.

4.3.1 Non-functional requirements

Table 4.2 gives the nine requirements and the way each is met.

Table 4.2: Non-functional requirements of the integrated platform

ID	Requirement	How it is addressed
NFR-01 Security	Access is authenticated and limited by role	Supabase Auth; role in profile; role checks in each service; row-level security on sensitive tables; secrets in host environment variables; SENTRY authentication added in the integrated application
NFR-02 Privacy	Collect and keep as little personal data as possible	SENTRY stores skeletons and not video and blurs faces; GLOSS stores no video; PULSE shows a synthetic display identity; SCRIBE was trained on de-identified text
NFR-03 Human control	Outputs are advisory, explained and reviewable	SHAP factors; review step for PULSE suggestions; deferral below 0.35 in GLOSS; records open to review in SCRIBE
NFR-04 Performance	Results arrive fast enough for the task	SENTRY classifier 5.55 ms per window on CPU; GLOSS model stage 73 to 144 ms; SCRIBE 51 s of audio in 12.2 s on a GPU and 77.2 s on a CPU
NFR-05 Usability	Simple for people with little time	One sign-in; role-based sidebar; short GLOSS sessions; spoken input in SCRIBE
NFR-06 Reliability	Degrade safely when something fails	GLOSS works without the language model; PULSE gives no flag before five assessments; SENTRY smooths scores over time; SCRIBE rules back up the extractor
NFR-07 Maintainability	Modules can change independently	Separate modules in one repository; table prefixes; 20, 49, 12 and 23 component tests
NFR-08 Scalability	Capacity can grow with the number of homes	Services scale separately; SCRIBE gains speed on GPU hosting; GLOSS needs a worker pool beyond about a hundred users
NFR-09 Compliance	Meet data protection law	Sri Lanka's Personal Data Protection Act, No. 9 of 2022, applies to staff and health data from 1 January 2027

Source: Compiled by the authors.

4.3.2 Security, privacy and ethical design

Social, security and ethical aspects were treated as design inputs and not as afterthoughts. Table 4.3 sets out the main risks of placing staff, resident and health-related data on one platform, the control chosen for each, and the further safeguard planned. The treatment follows the themes that recur in AI ethics guidelines [12].

Table 4.3: Security, privacy and ethical risks and controls

Aspect	Risk	Control in the design	Further safeguard planned
Unauthorized access	A user sees staff scores or resident events outside their role	Authenticated sessions; role and team scope in services; row-level security; audit log	Penetration test of the hosted platform before the pilot
Staff surveillance	Stress scores are used to judge or discipline caregivers	Scores visible to supervisors and administrators only; wording states that a score is not a diagnosis; advice from Remark Elder Care that a score should start a conversation	A signed use policy and caregiver consent in any pilot
Resident privacy	Video of residents is stored or leaked	Skeleton extraction on site; faces blurred; no raw video stored; GLOSS keeps no video	Site-specific review of the camera path at each home
Wrong or biased output	A false alarm, a missed fall or a wrong sign judgement	Temporal smoothing, deferral when unsure, rule backstops, human review, reported error rates	Calibration with care-home data during the pilot
Alarm fatigue	Staff learn to ignore alerts [58], [29]	Alert states with sustained evidence; acknowledgement; cooldown between HIGH alerts	Tuning of thresholds for each site
Data protection law	Unlawful processing of health and staff data	Data minimization; de-identified training data; consent before real monitoring	Compliance review before the PDPA duties apply on 1 January 2027
Research ethics	Collecting data from people without approval	Ethics application submitted; no identifiable data collected meanwhile	Collection under consent once the decision is received
Social impact	Over-reliance on automation; exclusion of staff without devices	Advisory outputs only; paper and spoken alternatives; training	Study of caregiver experience in the pilot

4.4 Integrated Workflow and Exception Handling

In normal operation a user signs in, the application reads the role and shows the modules that role may use, and each module works as described in Chapter 3. Table 4.4 lists the important failure paths and what the platform does. The module behaviors are unchanged in the integrated application, because the modules were not altered when they were joined.

Table 4.4: Important exceptions and the intended behavior

Situation	Behavior	Component
A user without the required role opens a module or calls an API	The service refuses the call with status 403; the module does not appear in the sidebar	All
Fewer than five scored assessments for a caregiver	No flag is raised and the caregiver is not considered further in the redistribution pipeline	PULSE
No same-unit candidate with a score below 0.35	No substitute is suggested; the supervisor decides	PULSE
A supervisor does not agree with a suggestion	The suggestion is recorded as reviewed; nothing is assigned automatically	PULSE
A movement that is brief or ambiguous	Temporal smoothing keeps the state at NORMAL or MODERATE; HIGH needs sustained evidence (about 2.5 s)	SENTRY
A long or multi-sentence voice note	Audio is cut into chunks of up to 30 s and sentences are processed separately	SCRIBE
A note that mentions a symptom or abnormal vital sign	Field-based alert rules raise an alert even if the extractor missed it	SCRIBE
Recognizer confidence below 0.35	The learner is asked to repeat; no correction is given and the language model is not called	GLOSS
Language model unavailable	Rule-based feedback is shown	GLOSS
A hosted service is unavailable	The failure is handled in the front end and in the service layer, so the application stays usable and the other modules keep working	Platform

4.5 Team Contribution and Dependency Matrix

Table 4.5 maps each member to the work carried out. Component research and development belongs to the member who owns the component, while the shared work covers the parts of the platform on which every module relies.

Table 4.5: Contribution of each member

Member	Component research and development	Shared platform and integration	Test evidence
Dayana Priyadharshani Kumar (IT22178640)	PULSE: data split, two-stage model, explanations, redistribution rule, 18 routes, supervisor views	Integration of the four components; deployment on Render and Vercel	20 PULSE test cases; held-out evaluation
Harishalinee Elangovan (IT22057488)	SENTRY: skeleton windows, ST-GCN and biomechanical fusion, alert states, dashboard, API and WebSockets (the server side of the alert link to the mobile client)	Security of SENTRY in the integrated application	49 pytest tests; dataset and webcam experiments

Member	Component research and development	Shared platform and integration	Test evidence
Vithursha Prabagaran (IT22294098)	SCRIBE: speech recognition, disfluency removal, grammar, ADL extraction, alert rules, handover	SCRIBE service and tables brought under the shared sign-in	12 SCRIBE test cases; evaluation sets
Kaushalya Nagenthraraja (IT22289384)	GLOSS: recognition ensemble, DTW feedback, guard, Parkinson's module, progress and quiz	gloss_ and pdedu_ schemas and migrations	23 GLOSS tests; held-out evaluation

Table 4.6 shows what each component needs from the shared parts. No module calls another module's service. The components meet only through the shared identity and through the supervisor who receives their outputs.

Table 4.6: Dependencies of each component

Component	Shared parts it needs	Depends on another component?	External dependency
PULSE	Auth, profiles, caregiver_profiles, audit_log, model_registry	No	Supabase; Render
SENTRY	Auth, caregiver_profiles (read-only), audit_log, R2	No	Camera and on-site computer; mobile client
SCRIBE	Auth, caregiver_profiles, audit_log, model_registry, R2	No	GPU hosting for speed; text-to-speech service
GLOSS	Auth, caregiver_profiles, audit_log, model_registry, R2	No	Optional language model service

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common Technology Stack and Infrastructure

Four students had to build, host and maintain one platform, and that fact shaped every technology choice. The server side uses Python and the browser side uses TypeScript, so the team works in two language families only. A managed database with built-in authentication gives all four modules one set of users and one set of access rules. Hosting is on free or low-cost tiers, which keeps a pilot affordable. The proposal had planned Flask with SQLite, but that pair could not offer a single sign-in and common policies across four modules, and it was replaced as the components came together. Table 5.1 lists the shared layers, the reason for each choice and the alternative that was set aside.

Table 5.1: Shared technology stack with justification and alternatives

Layer	Choice	Why it was chosen	Alternative set aside
Front end	React with TypeScript, built with Vite	One application and one component model for four modules; typed code catches interface errors early	A separate front end for each component
Back end	FastAPI services in Python	Same language as the models; typed routes; WebSocket support for SENTRY	Flask, used in the proposal
Database and authentication	Supabase: PostgreSQL, Auth and row-level security	Managed service with roles and policies, so access rules are written once	SQLite with hand-written login
File storage	Cloudflare R2	Low-cost object storage for large files kept out of the database	Files stored inside database rows
Hosting	Render for services; Vercel for the front end	Deployment straight from the repository; free or low-cost tiers for a pilot	A self-hosted server
Testing	pytest and written test cases	Repeatable checks that run on every change	Manual checking only

Each component then added the technology that its own problem called for. Table 5.2 gives the main choices with the alternatives that were tried or considered. Several of them were settled by measurement on validation data and not by preference. The

GLOSS ensemble was chosen on validation macro-F1, for instance, and the SENTRY alert logic dropped a zone modifier once it was seen to produce false HIGH states.

Table 5.2: Component-specific technology choices and alternatives

Component	Choice	Justification	Alternative considered
PULSE	Two-stage gradient-boosted trees [46] (LightGBM) with SHAP explanations [47]	Strong on small tabular data; a per-person expected heart rate separates a person's normal from their deviation; each score can be explained	Single-stage model; deep tabular networks
SENTRY	14-joint skeleton; ST-GCN [26] fused late with 18 biomechanical features	Small enough for a CPU; late fusion keeps the hand-built features interpretable; windows of 90 frames hold a whole fall	33 MediaPipe landmarks or 17 COCO joints; LSTM, GRU or Transformer; early fusion; 1 s or 5 s windows
SENTRY	Smoothed scores (EMA 0.25) with a HIGH gate of 0.72 held for 25 windows	Cuts false HIGH states by requiring sustained evidence	Raw window scores; zone modifier (removed)
SCRIBE	Whisper-small [49] with LoRA [50], rank 16	Adapts to accented speech on modest hardware	Larger Whisper models; training from scratch
SCRIBE	BART-base [51] for disfluency, T5-small [52] for 19-field extraction	Small sequence-to-sequence models with a measured exact match of 92.3%	T5-small for both stages; a large language model for extraction
SCRIBE	Templates for the handover report; a text-to-speech service for the spoken version	Templates cannot invent facts, which a generative model might [34]	Generated reports; a licensed speech service
GLOSS	TCN [53] and SPOTER-style transformer [40] ensemble	Chosen on validation macro-F1; better than the best single model by 0.038	Either model alone; ST-GCN
GLOSS	DTW [55] against a medoid reference sign	Accepts different signing speeds and gives interpretable errors	Fixed-length comparison; a 3D avatar guide
GLOSS	Rules write the feedback; a language model may only reword it behind a guard	Keeps corrections faithful to the measured error [56]	Free generation by a language model

5.2 Development and Integration Process

The team worked in two phases. In the first, each member built their own component against the plan and objectives agreed in the proposal. The Microsoft Planner board kept those plans in view. Its early buckets cover proposal and approval, literature review, methodology and system design, and data collection and preprocessing, while its later buckets cover model development, system development, integration, and

testing and validation (Figure 5.1 and Figure 5.2). Separate buckets held the research paper, the presentations and the report writing.

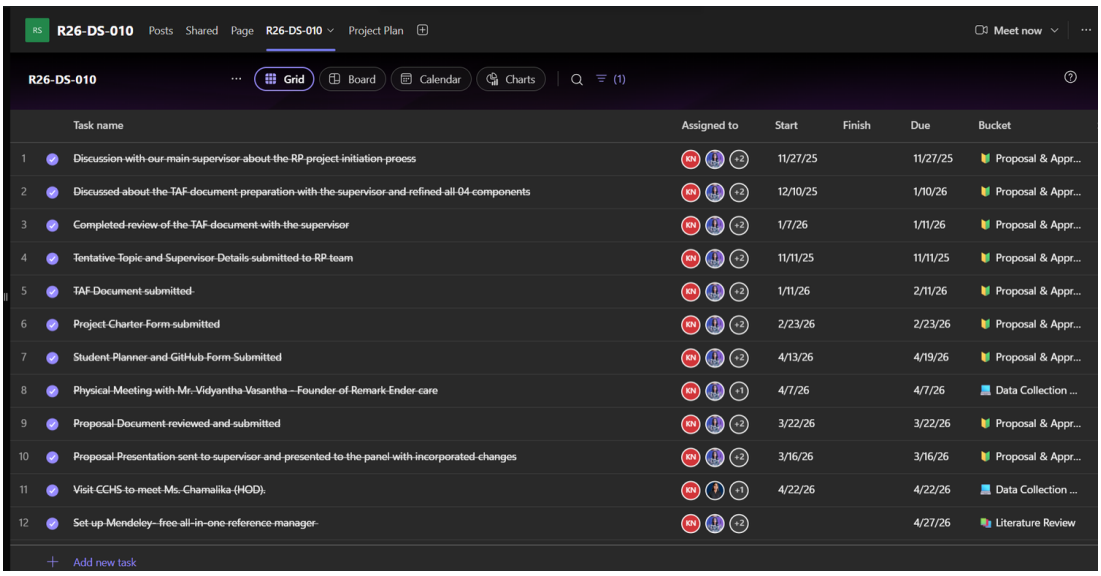


Figure 5.1: Planner grid of the completed proposal-stage tasks and their dates

Source: Screenshot of the group's Microsoft Planner board, 2026.

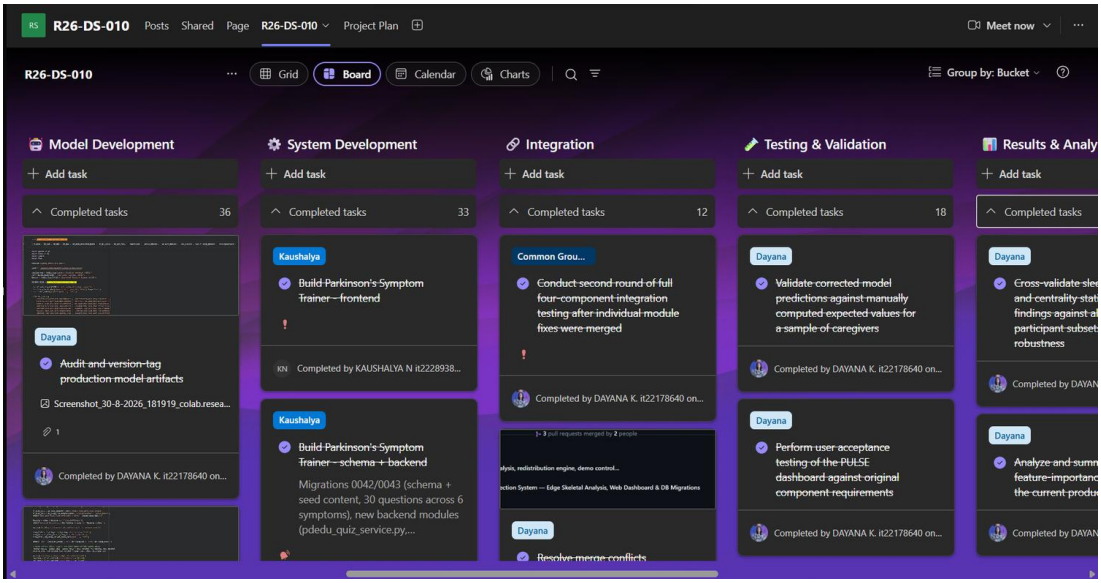


Figure 5.2: Planner buckets for development, integration and testing

Source: Screenshot of the group's Microsoft Planner board, 2026.

In the second phase the four components were joined in the shared application, as described in Section 4.1. The Planner records a second round of full four-component integration testing, carried out after the fixes to the individual modules had been

merged. That entry shows the round was completed, and Section 6.2 reports the integration checks that go with it.

All code is held in one public GitHub repository, R26-DS-010-Intelligent-Care-Support, which has four contributors (Figure 5.3). Its README describes the layout. The front end keeps one folder per module under src/modules, each service has its own folder under backend_services, and the database schema lives in versioned migrations managed with the Supabase command-line tool. Every member developed on a named branch: PULSE-01, SENTRY-03, SCRIBE-04 and GLOSS-02. The branches were then merged into the integration branch pp2-foundation. Figure 5.4 shows the repository structure, and Appendix D shows each branch.

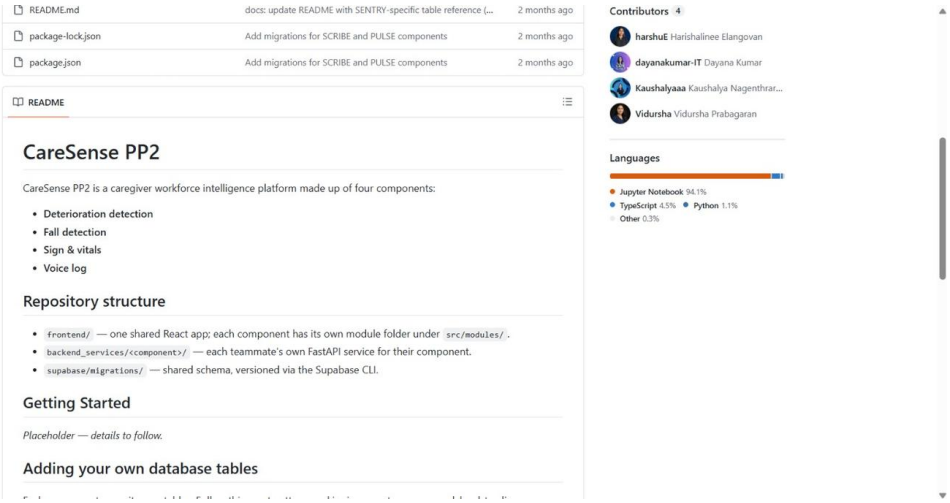


Figure 5.3: Repository page showing the README structure and contributors

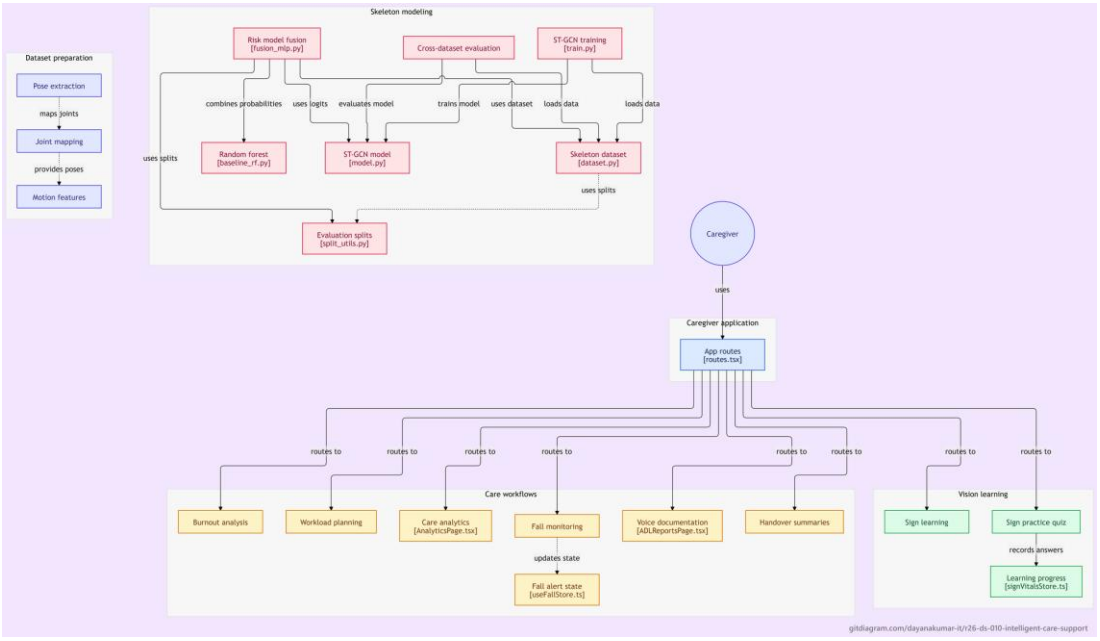
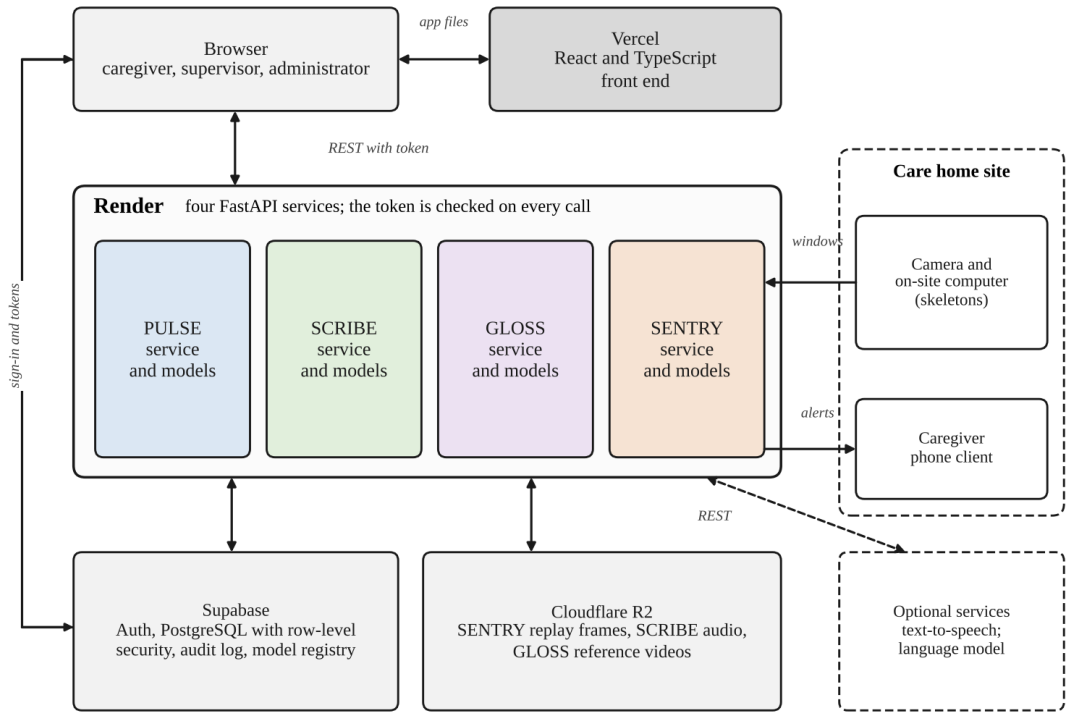


Figure 5.4: Repository structure generated from the code

5.3 Deployment and Operational Configuration

Figure 5.5 shows how the platform is deployed. The React front end is served from Vercel: the browser downloads the application bundle and runs the shared application and the four modules on the user's own device. The FastAPI services run on Render, one for each module. Every request carries the sign-in token, the service checks the user's role and team, and it refuses the call with a 403 status when the role is not permitted. Cross-origin settings admit only the address of the front end. Supabase provides authentication, by issuing the tokens that the services check, and the PostgreSQL database that holds the profiles, the caregiver identity, the audit log, the model registry and the tables of each module. Row-level security adds a second layer of access control inside the database. Cloudflare R2 is the object store for files too heavy for database rows, namely SENTRY replay frames, SCRIBE audio recordings and GLOSS reference videos. Each module writes to its own path, and the files are reached only through the services.



Dashed boxes lie outside the hosted platform. Only skeleton windows leave the care home; video is not sent to the hosted services.

Figure 5.5: Deployment of CareSense

No service holds a password in its code. The secret that matters most is the Supabase service-role key, because it lets a service skip the row-level security checks and read or write any record, staff and resident data included. The repository is public, so a key committed there would be open to anybody. Each key is therefore typed into the hosting settings on Render, and the running service picks it up when it starts. The browser bundle on Vercel receives only the public Supabase address and key, and what that key can reach is limited by the database policies for the signed-in role.

SENTRY runs in two places. A camera and a computer in the care home extract the skeletons locally, and only the skeleton windows leave the room, so video never reaches the hosted services. The on-site computer sends these windows over an authenticated HTTPS and WebSocket connection to the SENTRY service on Render. The service scores them and passes the results to the dashboard on Vercel and to the caregiver's phone.

Day-to-day operation relies on the standard facilities of the hosts. Render keeps the logs and the health status of each service, and the services expose health routes that show whether they are running. Supabase provides managed database backups, and the model files and migrations are held in the project repository. A failed service or database can therefore be restored from a backup and redeployed from the repository.

The audit log is the basis of accountability inside the platform. It records each significant action together with the identity of the user who performed it and the time at which it happened. When a stress score, a fall alert or a voice-note record is later disputed, the relevant entry can be retrieved and reviewed, and the matter is settled on documented evidence and not on recollection.

5.4 Evidence of Team Execution

Table 5.3 summarizes the evidence that the work was planned, supervised and reviewed over the year. The meeting dates in the table are those of the internal and supervisor meetings for which a screenshot is held, and each date can be checked against the Microsoft Teams records in Appendix C, from which the access links have been removed. They represent only part of the meetings that took place. The group also met on several other occasions, some in person and others online.

A stand-up meeting was held every week as well. Each member reported what had been completed, what was planned next and what was blocking progress, so problems between the components were raised early instead of surfacing at integration.

Microsoft Planner served as the working record of the group. Its tasks are organized into buckets that follow the stages of the project, with separate buckets for report writing, presentations and the research paper. Figure 5.1 and Figure 5.2 show how the tasks are laid out. A task was marked complete only once its output had been attached to it, whether that was a PDF document, a diagram produced for the stage or the slides prepared for the proposal and progress reviews. The Planner therefore shows both that a stage was finished and the work product that came out of it.

The same approach was followed for the visits to partner organizations. The Planner records the meeting at Remark Elder Care and the visit to the Colombo Caregiver & Health Science School as completed tasks, and the evidence for each visit is held with

them. Appendix C reproduces the signed visit letters and the photographs taken at the sites. The supervisor meeting records, the Planner buckets, the visit evidence and the repository history agree with one another.

Table 5.3: Evidence of team execution

Record	What it shows	Where to find it
Topic and supervision	Topic submitted 11th November 2025; supervisor discussion 27th November 2025; assessment form refined 10th December 2025, reviewed 7th January 2026 and submitted 11th January 2026	Planner grid; Teams invitations in Appendix C
Proposal stage	Project charter 23rd February 2026; proposal presentation 16th March 2026; proposal document 22nd March 2026	Planner grid
Stakeholder visits	Remark Elder Care 7th April 2026; CCHSS 22nd April 2026; repeated visits to Love and Care Elder Home	Visit letters in Appendix C; Table 1.2
Progress review	Group meeting 3rd May 2026; supervisor feedback meeting after the progress presentation on 11th May 2026	Teams records in Appendix C
Later work	Group meetings on 4th August and 21st August 2026 (research progress; conference paper drafting) and 8th September 2026 (journal paper writing)	Teams records in Appendix C
Version control	One GitHub repository with four contributors; one branch per component merged into pp2-foundation	Section 5.2; Appendix D
Data approvals	NTU RGB+D access approved for SENTRY on 17th June 2026; TILES-2018 data use agreement signed in July 2026	Appendix C

Source: Group's Microsoft Planner and Teams records; visit letters; approval correspondence.

6 INTEGRATED TESTING, EVALUATION AND VALIDATION

6.1 Evaluation Framework and Success Criteria

The evaluation asks four questions at group level. First, does each component work well enough, on data it has not seen, to serve as decision support? Second, do the components work together as one platform? Third, are access and privacy protections in place? Fourth, do the intended users find the platform useful and acceptable? Table 6.1 gives the criterion used for each.

Table 6.1: Evaluation framework

Question	Acceptance criterion	Metrics and data	Analysis
EQ1 Component performance	Held-out results reported with uncertainty and compared with a baseline; limits stated	ROC AUC, sensitivity, false alarm rate, word error rate, exact match, macro-F1; data separated by person, subject or signer	Confidence intervals; baselines; error analysis for each component
EQ2 Integration	All four modules open from one sign-in and keep their own routes, tables and models	Integration checks (Section 6.2)	Comparison with the shared contracts
EQ3 Security and privacy	Role checks, authentication and data minimization present in every module	Role-based access tests; review of stored data	Comparison with Section 4.3
EQ4 Stakeholder acceptance	Intended users have seen the components and their comments are recorded and acted on	Walkthroughs, demonstrations, feedback records	Qualitative summary; planned user study

6.2 Integration and End-to-End Testing

6.2.1 Platform integration checks

Table 6.2 lists the integration checks that matter for the whole platform and the evidence for each.

Table 6.2: Integration checks and the evidence for each

Check	What is checked	Evidence	Status
Single sign-in and role-based sidebar	A user signs in once and the modules appear according to role	Role-access test of the PULSE service; integrated screenshots (Figure 3.1 and Figure 3.2)	Verified
Role scope	A supervisor sees only their own team in PULSE	Two PULSE test cases on team scope	Verified

Check	What is checked	Evidence	Status
Shared model registry	The registry lists a model for each module that has one	Active model shown on the PULSE overview	Verified
Shared caregiver identity	The same caregiver_profiles record serves all four modules	The same caregiver appears in the SENTRY and Voice Log screens	Verified by use
Module independence	Adding modules did not change PULSE routes, tables or models	Repository history and PULSE test results	Verified
SENTRY secured in the integrated application	API and WebSocket reject unauthenticated use	Security test carried out on the integrated application by the SENTRY owner	Confirmed by the group
Full four-component integration test	Second round after module fixes were merged	Completed task in Planner (Figure 5.2)	Completed
Hosted service failure	A service that is down does not break the application	Group confirmation	Confirmed by the group
Performance on Render	Response time of each module in the hosted setting	Component timings on CPU and GPU (Section 4.3.1)	Planned for the pilot
User acceptance of the whole platform	Users complete tasks in the integrated application	PULSE walkthrough with a caregiver and a Remark Elder Care representative	Planned for the pilot

6.2.2 Component test coverage

Each component was tested before it joined the platform, and the tests were chosen to show that it can operate inside the shared environment, not only on its own.

The PULSE service was checked with 20 test cases. They cover role-based access, including a supervisor requesting a caregiver outside their team and a non-administrator calling an upload route. They also cover scoring with missing features, the flagging rules (sustained risk with and without a candidate, fewer than five assessments, and duplicate recommendations), the supervisor's review step, rejection of an invalid upload with a 400 response, and a service start-up that fails when the model file is missing.

SENTRY has 49 automated tests, and all of them passed at the end of the project. They cover feature extraction, the post-processing that turns window scores into alert states, the shapes of the model's inputs and outputs, and the REST endpoints. A separate security test was carried out on the integrated application to confirm that its API and WebSocket reject unauthenticated use.

The 12 SCRIBE test cases cover recordings longer than 30 seconds, input containing several observations, the consistency between the written and the spoken handover, and identical output on a CPU and a GPU, which gave the same 9 records and 2 alerts. Three cases failed at first, for audio decoding, the parser for the 19-field output and the spoken handover text, and they passed once the defects were corrected. The case with three observations in one input now produces three correct records.

All 23 GLOSS tests passed. They cover authentication and the rule that the caller's identity is taken from the token and not from the request body, recognition of correct and wrong signs, the repeat request at low confidence, behavior when the language model fails, the guard on reworded feedback, the quiz mode when the camera is unavailable, and the Parkinson's quiz. Further cases confirm that the application produces the same prediction as the stored test model on ten stored cases and that the front end builds without type errors.

At platform level, a second round of full four-component integration testing was carried out after the individual module fixes had been merged. The shared sign-in, the role-based sidebar, the caregiver identity and the model registry were all checked in the integrated application.

6.2.3 Requirement-to-test traceability

Table 6.3 links each platform requirement from Section 4.3 and Section 3.1.2 to the test evidence that supports it. A requirement counts as verified only when the evidence is a recorded test or a held-out measurement.

Table 6.3: Traceability of platform requirements to verification

Requirement	Verification	Scenarios covered	Result
FR-01, FR-02	PULSE and GLOSS service tests; SENTRY authentication test	Valid role returned; call outside the team; non-administrator on an upload route; request without a valid token rejected; caller identity taken from the token and not the request body	Passed
FR-03, FR-04	Integrated screens; registry on the PULSE overview	Same caregiver in two modules; model listed	Verified by use
FR-05	PULSE scoring tests; held-out evaluation	Score with and without missing features; unscored days; ROC AUC 0.7211	Passed

Requirement	Verification	Scenarios covered	Result
FR-06	PULSE flagging and review tests	Sustained risk with candidate; no candidate; fewer than five rows; duplicate; review	Passed
FR-07 to FR-09	SENTRY 49 pytest tests; unseen-subject evaluation	Feature extraction, post-processor, model input and output shapes, REST endpoints; sensitivity 92.31%, false alarm rate 1.28%	Passed
FR-10, FR-11	SCRIBE 12 test cases; held-out and safety sets	Long recordings, several observations, chest-pain alert, report and speech consistency, CPU and GPU parity	Passed after fixes
FR-12, FR-13	GLOSS service tests; unseen-signer evaluation	Correct and wrong sign, hedged message, low-confidence repeat, threshold tiers, mastery, next lesson, execution score; macro-F1 0.7752	Passed
FR-14	GLOSS quiz test	Ten-question Parkinson's quiz with score and explanations; educational wording	Passed
FR-15	Users and Roles screen of SENTRY	Room assignment by caregiver	Verified by use
NFR-01 to NFR-03	Component tests; design review (Section 4.3.2)	Role checks; privacy design; explanations and deferral	Verified
NFR-04	Component timings	Per-stage latency on CPU and GPU	Verified per component
NFR-05, NFR-08	PULSE walkthrough; one hosted service per module	Prototype demonstration to a caregiver; services deployed and scaled separately	Demonstrated
NFR-06	GLOSS and PULSE tests	Language model failure; camera unavailable; fewer than five rows; model file missing	Passed
NFR-07	Repository structure; component tests	Module separation; 104 component tests across the four modules	Verified
NFR-09	Review of data handling against the Personal Data Protection Act	Duties for health and staff data identified	Duties identified

6.3 Overall Results

Table 6.4 brings together the principal held-out result of each component so that the four can be compared side by side.

Table 6.4: Principal held-out results of the four components

Component	Principal result	Comparison	Reading of the result
PULSE	ROC AUC 0.7211 (95% CI 0.6644 to 0.7699), 27 unseen caregivers, 1,406 assessments	Grouped cross-validation mean 0.6991; without the previous response 0.6108; gain over the previous-response baseline +0.0247	Moderate discrimination from wearable data alone, on hospital workers who were never seen in training
SENTRY	Unseen NTU subjects: accuracy 97.12%, sensitivity 92.31%, false alarm rate 1.28%, AUC 0.9836	NTU to UR AUC 0.410; fine-tuned on webcam clips AUC 0.8527 (93 clips, two volunteers)	Strong on the benchmark, and a short fine-tune restores performance in a new setting
SCRIBE	Word error rate 15.57%; T5 exact match 92.3%; alert recall 93.3% with rules on the safety set	Word error rate before adaptation 26.13%; model-only alert recall 82.0%	Adaptation cuts speech errors by more than a third, and the rules add eleven points of recall
GLOSS	Macro-F1 0.7752 (95% CI 0.712 to 0.802), accuracy 0.7859 on unseen signers	+0.038 over the best single model; confident predictions 98.4% correct	Dependable when the model is confident, and uncertain attempts are sent back for repetition

The four components were each evaluated on people, subjects or signers that were set aside before development began, and each one performed well enough to support the task it was built for.

The modest size of some results does not weaken the platform, because of the way the outputs are used. Each component presents its result as advice to a person. A low-confidence sign is sent back for a repeat attempt, a stress suggestion waits for the supervisor's review, and a missed symptom is caught by a rule. In this setting a dependable, explainable result is worth more than a very high score, and the platform is built to give the first. Considered as a whole, the results show a decision-support system whose parts work, whose limits are known and whose next step is validation in a real care home.

6.4 Stakeholder and User Validation

Each component of the platform has been put in front of the people it is designed to serve, and the response has been constructive throughout. Table 6.5 sets out the feedback by component and the refinement each comment produced.

Table 6.5: Stakeholder feedback on the components and resulting refinements

Component	Who	Feedback	Refinement
PULSE	Remark Elder Care representative; an elder-care caregiver (prototype walkthrough, Aug. 2026)	A suggested substitute should reflect skill mix, shift pattern and minimum staffing; the score should start a conversation; caregivers should consent before monitoring	Recorded as requirements for the pilot and reflected in the wording of the module
SENTRY	Staff at Love and Care Elder Home and Remark Elder Care (Aug. 2026)	Welcomed the intended functions and the value of prompt, reliable alerts	Alert design kept simple, with three states and acknowledgement
SCRIBE	Staff of two care homes; a caregiver who reviewed the 200 safety-set labels; caregivers who tried the system	Positive response; questions about long, multi-sentence recordings	Audio chunking and sentence splitting added
GLOSS	Caregiver trainees and a sign-language tutor (CCHSS); reviewers; Remark Chairperson	Skills fade after training; recognition alone is not enough; different signing speeds should be accepted	Fixed 59-sign vocabulary with repeat until mastered; DTW scoring; Parkinson's module; caregiver study protocol prepared

Source: Compiled by the authors from stakeholder discussions.

For PULSE, the walkthrough with a Remark Elder Care representative and an elder-care caregiver was well received. Both found the idea of a score that opens a conversation about a caregiver's wellbeing practical. Their suggestions on skill mix, shift pattern, minimum staffing and caregiver consent now shape how the module is configured and how it is introduced. SENTRY was discussed with staff at Love and Care Elder Home and Remark Elder Care, who welcomed its intended functions and saw clear value in prompt, reliable fall alerts for residents.

SCRIBE drew a positive response from the staff of two care homes and from the caregivers who tried it. A caregiver also reviewed the labels of the 200-case safety set, which strengthened the reference against which the alerts are measured. The questions raised about long, multi-sentence recordings led directly to audio chunking and sentence splitting, so the module now handles extended spoken notes with ease.

GLOSS received strong support from caregiver trainees, a sign-language tutor from CCHSS and the Remark Chairperson. Their emphasis on skills fading after training and on accepting different signing speeds gave rise to the fixed 59-sign vocabulary, DTW-based scoring and the Parkinson's education module. A caregiver study protocol has also been prepared, which gives the module a clear route into formal evaluation with trainees.

6.5 Achievement of Common Objectives

Table 6.6 measures each group objective against the evidence in Sections 6.1 to 6.4. For each one it names the outcome that was intended, cites the test results, traceability links, stakeholder feedback or deployment records that support it, and gives the resulting status. An objective counts as achieved only where this evidence demonstrates the outcome it describes.

Table 6.6: Achievement of group objectives

No.	Status	Justification
GO1	Achieved at the level of the application	PULSE scores caregiver strain with explained factors and a reviewable suggestion. Testing on 27 unseen caregivers gave an ROC AUC of 0.72, and the module is part of the integrated application.
GO2	Achieved at the level of the application	SENTRY detects falls from skeletons, stores no video and reaches 92.31% sensitivity at a 1.28% false alarm rate on unseen subjects. A short fine-tune on webcam clips brought the cross-setting score to 0.85.
GO3	Achieved at the level of the application	SCRIBE turns a spoken note into a structured 19-field record with alerts. Alert recall rises from 82.0% to 93.3% once rules are added, and the written and spoken handovers agree.
GO4	Achieved at the level of the application	GLOSS recognizes 59 care signs on unseen signers, asks for a repeat when unsure and gives explained feedback. The Parkinson's module adds education on early motor signs, and the caregiver study protocol is ready.
GO5	Achieved at the level of the application	The four modules run in one application with shared sign-in, roles, identity, audit log and registry. The integration checks in Section 6.2 confirm that they operate together.
GO6	Achieved at the level of the application	Every module keeps a person in control, limits the data it holds and defers when unsure. Role checks, minimization and the audit log are in place across the platform (Section 4.3.2).
GO7	Achieved at the level of the application	Results are reported on held-out data with intervals and limits, and Chapter 7 sets out a costed commercial route for care homes.

6.6 Integrated Limitations and Threats

Knowing where a result applies is part of using it well. The points below mark the boundaries of the present evidence and show where the next phase of work will extend it.

- **Data domain.** Every model was trained on data from other settings: hospital workers, laboratory skeleton data, public speech and public sign data. Performance in care homes may differ, and the cross-dataset result for SENTRY shows how large the difference can be. Site-level calibration during the pilot addresses this directly.
- **Shared points of failure.** The shared sign-in, database and storage serve every module. SENTRY also depends on equipment on site and SCRIBE on a GPU for fast processing. Separate services and health routes limit the effect of a single failure.
- **Labels and thresholds.** PULSE labels are self-reported ratings, SENTRY thresholds were set on small webcam data, SCRIBE's training text is de-identified and not care-home audio, and GLOSS recognizes 59 signs in one sign language. The estimates for PULSE and GLOSS have wide intervals, which larger samples in the pilot will narrow.
- **Generalizability.** The platform was designed for small care homes in Sri Lanka, and the pilot is the step that extends it to other homes and settings.
- **Operating conditions.** Behavior under load, the camera path for SENTRY, platform-level security testing and monitoring will be demonstrated on the hosted platform as part of pilot preparation.

7 COMMERCIALISATION, SUSTAINABILITY AND RESPONSIBLE INNOVATION

This chapter treats CareSense as a single product: a bundle of the four modules offered to care homes on subscription. The market facts come from the stakeholder visits and the sources cited. The prices are proposals, built from published hosting prices and the weight of each component, and the pilot described in Section 8.3 is where customers will respond to them.

7.1 Value Proposition and Beneficiaries

To a care home, the proposition is simple. One subscription supports the four things that fill a shift: knowing how staff are coping, noticing falls, recording care and communicating with residents. Table 7.1 sets out the value for each beneficiary.

Table 7.1: Beneficiaries and the value created

Beneficiary	Value	Basis
Care home owners and managers (buyer)	Earlier notice of strain, complete and searchable records, fewer missed events, support for staff retention, evidence of good record keeping for inspection	Visits to Remark Elder Care and Love and Care Elder Home; component results
Supervisors	One view of staff strain, alerts and handovers; suggestions that reflect unit, skill mix and staffing	PULSE walkthrough; SCRIBE and SENTRY designs
Caregivers	Less paperwork; a way to keep sign language skills; early warning of problems	Visits; CCHSS feedback; SCRIBE and GLOSS designs
Residents and families	Faster response to falls; fewer missed observations; better communication	Design intent, to be measured in the pilot
Training schools (partner)	Practice that continues after the course	CCHSS visit

7.2 Market Alternatives and Differentiation

The first target segment is small and medium elder-care homes in Sri Lanka, the setting in which the stakeholder visits took place. Training schools and home-care services form secondary segments. People preparing for caregiver work abroad are a further audience for GLOSS, since a shared sign language is a practical skill in any care setting.

Table 7.2: Alternatives to CareSense and the opening they leave

Need	Alternatives	Opening for CareSense
Fall detection	Bed and chair alarms, wearables, and depth or thermal room sensors [20], [22]	A camera with local skeleton processing sits between contact devices and costly sensors
Documentation	Handwritten files and typed record systems [9]	Spoken notes turned into ADL fields, alerts and accent-adapted recognition for care homes
Sign language and symptom awareness	General sign-language apps and classroom courses [41]	A care vocabulary with explained feedback and a view for the institution
Staff strain	Supervisor observation and self-report [2]	A regular, explainable indication with a suggested response

The bundle stands apart in four ways. First, it is designed for the care home and not for the hospital or the home user. Second, it places four needs behind one sign-in, so a home runs one system instead of four. Third, it keeps people in control and explains what it produces. Fourth, it holds costs down by using small models and existing equipment. Testing these advantages in the market is the purpose of the pilot planned in Section 8.3.

7.3 Business Model and Feasibility

The proposed model is a monthly subscription for each care home, priced by the number of caregivers, with all four modules included. SENTRY needs a camera and a computer in every monitored room, so its hardware is leased as an add-on for each room. A one-time onboarding fee of Rs 40,000 covers set-up, staff training and device checks. A free three-month pilot in one care home, ending with a letter of intent, is the first commercial step.

7.3.1 Running cost and component weight

Prices were derived from cost. Table 7.3 lists the monthly running cost of the hosted platform at the published prices of Supabase, Render, Cloudflare R2 and Vercel, converted at Rs 330.61 per USD, the rate for 8 October 2026. Shared services, namely the Supabase Pro plan, the Render workspace, Vercel and storage, are divided equally among the four modules, since every module uses them. Each module then carries the cost of its own service. SCRIBE is the heaviest, because speech recognition needs a

larger instance (2 CPU and 4 GB) and would benefit from a GPU for fast turnaround, which can be priced once the pilot confirms the load. Part-time maintenance and support, estimated at Rs 60,000 a month, is also divided equally.

Table 7.3: Monthly running cost of the bundle and its division among modules (Rs)

Item	Basis	Rs per month
Supabase Pro (database, authentication)	USD 25	8,300
Render workspace	USD 25	8,300
PULSE service (1 CPU, 2 GB)	USD 25	8,300
SENTRY service (1 CPU, 2 GB)	USD 25	8,300
GLOSS service (1 CPU, 2 GB)	USD 25	8,300
SCRIBE service (2 CPU, 4 GB)	USD 85	28,100
Vercel Pro (one seat)	USD 20	6,600
Cloudflare R2 storage	Allowance of USD 20; R2 is priced per GB-month	6,600
Language-model rewording (GLOSS)	Estimate by the authors	1,000
Domain and e-mail	Estimate by the authors	1,500
Platform total		**85,300**
Part-time maintenance and support	Estimate by the authors	60,000
Fixed cost to recover		**145,300**

Source: Public price lists of Supabase, Render, Cloudflare and Vercel, accessed Oct. 2026; calculation by the authors.

Table 7.4: Share of the fixed cost carried by each module (Rs per month)

Module	Own service and add-ons	Share of shared services	Share of maintenance	Full cost
PULSE	8,300	7,825	15,000	31,125
SENTRY	8,300	7,825	15,000	31,125
SCRIBE	28,100	7,825	15,000	50,925
GLOSS	9,300	7,825	15,000	32,125
Total	**54,000**	**31,300**	**60,000**	**145,300**

Source: Calculation by the authors from the cost table.

7.3.2 Proposed tiers and break-even

Table 7.5 proposes three tiers. SCRIBE alone accounts for about a third of the platform cost, which is why a bundle priced at the level of a single-module product would not

recover its cost. The prices are set so that three average customers cover the fixed cost. The room kit for SENTRY is charged separately because it grows with the number of monitored rooms and not with the number of caregivers.

Table 7.5: Proposed bundle tiers and add-ons (Rs)

Tier	Who it is for	Price per month	Per caregiver
Starter	Care home with up to 10 caregivers	30,000	3,000
Standard	Up to 30 caregivers	55,000	1,833
Large	Up to 60 caregivers	85,000	1,417
SENTRY room kit	Camera and on-site computer, leased per monitored room	4,000 per room	Not applicable
Onboarding	Set-up, staff training and device check (one time)	40,000	Not applicable

Source: Proposal by the authors.

Table 7.6 shows how many homes are needed to cover the fixed cost of Rs 145,300. A single Large home brings in 85,000, two cover the cost, and a mix of one Large, one Standard and one Starter home brings in 170,000. The product can therefore break even with very few customers when hosting stays near the prices above. GPU hosting for SCRIBE and the room kits will add to the fixed cost, and the pilot will show by how much.

Table 7.6: Break-even in number of care homes

Customer mix	Monthly income (Rs)	Covers Rs 145,300?
5 Starter homes	150,000	Yes
3 Standard homes	165,000	Yes
2 Large homes	170,000	Yes
1 Large, 1 Standard and 1 Starter home	170,000	Yes
2 Standard and 1 Starter home	140,000	Close; one more Starter home covers it

Source: Calculation by the authors from the proposed tiers.

Feasibility has a technical and an operational side. Technically the platform runs and its components work on held-out data. Operationally, adoption needs a champion in the home, staff consent, a camera plan for SENTRY and time for training. Remark

Elder Care, Love and Care Elder Home and CCHSS gave access and advice during the project, and they are natural partners for the first pilot.

7.4 Adoption, Intellectual Property and Compliance

Adoption. Adoption depends on trust in the alerts, on assurance that stress scores will not be used to judge staff, on privacy where a camera is used, on stable internet and equipment, and on the effort of moving away from paper habits. The Remark Elder Care representative advised that PULSE scores should start a conversation and not serve as a judgement, and that caregivers must be told, and must consent, before real monitoring starts. These conditions belong in every customer contract.

Intellectual property. Each member holds the code, models and data resources that they produced, such as the ADL schema and dataset built for SCRIBE. The combined platform and the CareSense name are shared by the group. The platform also rests on third-party software and data. The terms of NTU RGB+D [48], WLASL [38], ASL Citizen [39] and the TILES-2018 data use agreement [16] will be checked before any paid use, since some of these resources were released for research. The text-to-speech service used by SCRIBE will be replaced by a licensed one for commercial use. Patent and trademark protection can be considered once the pilot confirms the commercial route.

Data protection and ethics. Sri Lanka's Personal Data Protection Act, No. 9 of 2022, treats health data as a special category, and its main provisions take effect on 1 January 2027. A commercial release will therefore need consent, access control, retention rules and support for the rights of data subjects, and the design already moves in that direction through minimization and role-based access. The ethics application for the project has been submitted, and identifiable care-home data will be collected only after approval is granted.

Safety and regulation. The platform is decision support and makes no clinical claim. A future version of SENTRY sold as a fall-detection function may come under medical-device rules, and it would then follow the software life-cycle and risk-management standards that apply to such products. The Parkinson's module in GLOSS

is educational and makes no diagnosis. A legal and regulatory review will precede any sale.

7.5 Sustainable Development Goal Contribution

CareSense contributes mainly to SDG 3, through the wellbeing of caregivers and the quality of care that older adults receive [5], and to SDG 8, through safer and lighter work for caregivers. GLOSS adds links to SDG 4 (skills) and SDG 10 (inclusion of deaf people), and SENTRY and SCRIBE add links to SDG 9 (innovation). The targets are listed in Table 1.4. These contributions are intended outcomes, and the pilot will measure them.

A responsible product also looks at its possible side effects. Continuous monitoring can feel like surveillance, alerts that come too often can lead to alarm fatigue [58], and a stress score can be misused for discipline. A wrong score or recognition result can give false reassurance, and caregivers without devices or reliable internet could be left out. Cameras and computers carry an energy and manufacturing footprint as well. The measures that protect the benefits are already part of the design and the contract: human review, deferral when unsure, consent, limits on who can see staff data, cautious alert thresholds and a review of results at each pilot site.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall Conclusion

Caregivers in care homes work under strain. They must keep residents safe, keep accurate records and communicate with residents who cannot rely on speech, while the tools that could help are separate and rarely designed for care homes. The group answered with CareSense, a single platform of four modules. Three conclusions follow from the evidence.

First, the problem can be tackled within one system. The four modules run in one application with one sign-in, shared roles, a shared caregiver identity, an audit log and a model registry, and they stayed independent when joined. Second, each module works on data it has not seen. PULSE ranks caregivers by stress risk with moderate accuracy, SENTRY detects falls well on benchmark subjects, SCRIBE lowers speech word error from 26% to 16% and extracts records with alert rules, and GLOSS recognizes 59 care signs and gives cautious feedback. Third, the platform is ready for the next stage. With ethics approval and partner care homes, CareSense can move into a pilot and be judged by the people it was built for. It is a credible, carefully controlled prototype of decision support for care homes.

8.2 Principal Integrated Contribution

The team produced three things of lasting value beyond the four components. One is a working integration pattern: a shared application, a role model and a few common tables that let four independent research components share a product without changing each other. Another is a common design stance for AI in care. Every output is advisory, uncertain outputs are deferred, explanations accompany scores that could be misread, and rules back up the models. The third is an open evaluation record in which each result is reported with its data set, its uncertainty and its limits, including the cross-dataset result for SENTRY and the modest gain from the added PULSE features.

8.3 Future Directions

Table 8.1 lists the work that would most strengthen the evidence and move the platform towards use, in order of priority.

Table 8.1: Priorities for future work

Priority	Work	Reason
1	Obtain the ethics decision and agree consent procedures with the partner care homes	Needed before any real data are collected
2	Show the integrated platform to supervisors, caregivers and administrators, and record their comments	Extends the component walkthroughs of Section 6.4 to the whole platform
3	Run a recorded end-to-end test and a security test of the hosted platform; document monitoring, backup and the SENTRY camera path	Builds on the integration checks of Section 6.2
4	Pilot in a care home for three months	Gives real use data and stakeholder feedback
5	Collect care-home data under consent: caregiver wearable data (PULSE), room recordings (SENTRY), caregiver speech (SCRIBE)	Tests every model in its real setting
6	Run the planned GLOSS study with caregivers	Shows whether learning improves
7	Calibrate SENTRY for each site	Cross-setting performance is the main risk
8	Review licenses, agree ownership and test prices with customers	Needed before any sale
9	Extend vocabulary and languages (for example Sri Lankan Sign Language) and add a licensed text-to-speech service	Local fit and commercial use

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APPENDIX A: INTEGRATED INTERFACE SPECIFICATION

Table A.1 lists the routes of each service, written as method and path. Table A.2 lists the shared tables.

Table A.1: Service routes of the four components

Service	Routes
PULSE	GET /health, GET /me; GET /caregivers, GET /caregivers/{id}/history, /risk-history, /baseline-history; POST /caregivers/{id}/simulate; GET /analytics/risk-summary, /risk-heatmap, /team-trends, /model-performance; POST /analytics/generate-redistribution-recommendations; GET /analytics/redistribution-recommendations; POST /analytics/redistribution-recommendations/{id}/mark-reviewed; POST /admin/upload-raw-data/{id}, /admin/upload-survey-responses/{id}, /admin/reveal-next-day/{id}; GET /admin/staging-status/{id}
SENTRY	GET /api/health, /api/system/status, /api/dashboard/summary; GET and POST /api/rooms; POST /api/rooms/{room_id}/camera; GET and PUT /api/rooms/{room_id}/zones; PATCH /api/rooms/{room_id}/caregiver; GET and POST /api/patients; GET /api/patients/{id} and /history; GET /api/alerts; PATCH /api/alerts/{id}/acknowledge; GET /api/events/{alert_id}/replay; GET /api/replay/status; GET /api/caregivers; POST /api/analyse; DELETE /api/alerts/clear-demo; GET /video/{room_id}; WebSocket /ws/live/{room_id}; WebSocket /ws/alerts
SCRIBE	Route groups for recording upload and processing, recording history, dashboard summary, handover brief and spoken handover
GLOSS	POST /gloss/attempts; GET /gloss/next-lesson; GET /gloss/signs/{sign_id}/reference; Parkinson's quiz endpoints under the pdedu_ module

Source: Service route listings of the four components.

Table A.2: Shared tables

Table	Purpose	Used by
profiles	Authenticated user and role (caregiver, supervisor, administrator)	All
caregiver_profiles	Caregiver identity linked to PULSE, GLOSS, SENTRY and SCRIBE	All
audit_log	Shared append-only event record	All
model_registry	Component, model name, version, training time, metrics and artifact paths	PULSE, SCRIBE, GLOSS

APPENDIX B: TEAM CONTRIBUTION DECLARATION

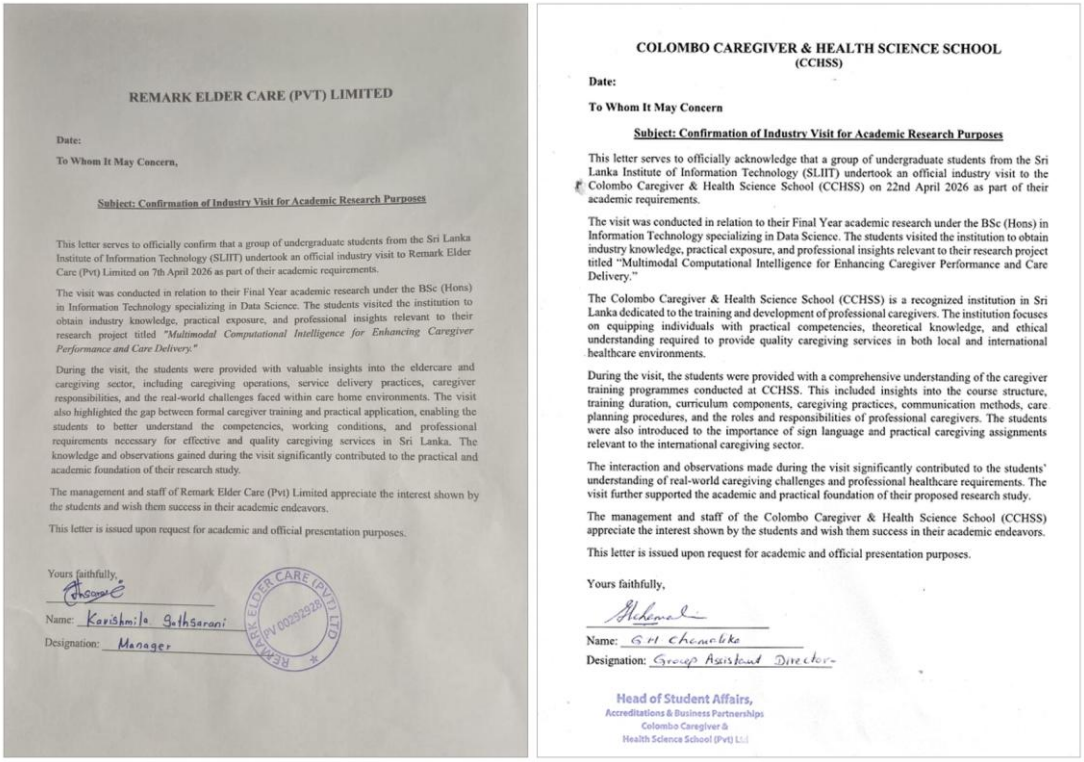
The record below states the contribution of each member. Each member confirms it by signing before submission.

Table B.1: Contribution declaration

Member	Individual contribution	Shared contribution
Dayana Priyadharshani Kumar (IT22178640)	PULSE: stress-risk model, explanations, redistribution rule, supervisor views	Integration of the four components; deployment
Harishalinee Elangovan (IT22057488)	SENTRY: skeleton-based fall detection, dashboard, API, WebSockets	Security of SENTRY in the integrated application
Vithursha Prabagaran (IT22294098)	SCRIBE: voice-to-ADL pipeline, alert rules, handover	SCRIBE service and tables brought under the shared sign-in
Kaushalya Nagenthraraja (IT22289384)	GLOSS: sign recognition, feedback, Parkinson's module, progress and quiz	gloss_ and pdedu_ schemas and migrations

APPENDIX C: EVIDENCE OF ENGAGEMENT, DATA APPROVAL AND PROJECT MANAGEMENT

This appendix reproduces the records that support Chapter 1 and Section 5.4. Access links and passcodes in meeting records have been removed, and the e-mail address on the data use agreement page has been masked.



REMARK ELDER CARE (PVT) LIMITED

Date:

To Whom It May Concern,

Subject: Confirmation of Industry Visit for Academic Research Purposes

This letter serves to officially confirm that a group of undergraduate students from the Sri Lanka Institute of Information Technology (SLIIT) undertook an official industry visit to Remark Elder Care (Pvt) Limited on 7th April 2026 as part of their academic requirements.

The visit was conducted in relation to their Final Year academic research under the BSc (Hons) in Information Technology specializing in Data Science. The students visited the institution to obtain industry knowledge, practical exposure, and professional insights relevant to their research project titled *"Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery."*

During the visit, the students were provided with valuable insights into the eldercare and caregiving sector, including caregiving operations, service delivery practices, caregiver responsibilities, and the real-world challenges faced within care home environments. The visit also highlighted the gap between formal caregiver training and practical application, enabling the students to better understand the competencies, working conditions, and professional requirements necessary for effective and quality caregiving services in Sri Lanka. The knowledge and observations gained during the visit significantly contributed to the practical and academic foundation of their research study.

The management and staff of Remark Elder Care (Pvt) Limited appreciate the interest shown by the students and wish them success in their academic endeavors.

This letter is issued upon request for academic and official presentation purposes.

Yours faithfully,

Name: Karishmila Gothsarani

Designation: Manager

REMARK ELDER CARE (PVT) LTD PV 00292928

COLOMBO CAREGIVER & HEALTH SCIENCE SCHOOL
(CCHSS)

Date:

To Whom It May Concern

Subject: Confirmation of Industry Visit for Academic Research Purposes

This letter serves to officially acknowledge that a group of undergraduate students from the Sri Lanka Institute of Information Technology (SLIIT) undertook an official industry visit to the Colombo Caregiver & Health Science School (CCHSS) on 22nd April 2026 as part of their academic requirements.

The visit was conducted in relation to their Final Year academic research under the BSc (Hons) in Information Technology specializing in Data Science. The students visited the institution to obtain industry knowledge, practical exposure, and professional insights relevant to their research project titled "Multimodal Computational Intelligence for Enhancing Caregiver Performance and Care Delivery."

The Colombo Caregiver & Health Science School (CCHSS) is a recognized institution in Sri Lanka dedicated to the training and development of professional caregivers. The institution focuses on equipping individuals with practical competencies, theoretical knowledge, and ethical understanding required to provide quality caregiving services in both local and international healthcare environments.

During the visit, the students were provided with a comprehensive understanding of the caregiver training programmes conducted at CCHSS. This included insights into the course structure, training duration, curriculum components, caregiving practices, communication methods, care planning procedures, and the roles and responsibilities of professional caregivers. The students were also introduced to the importance of sign language and practical caregiving assignments relevant to the international caregiving sector.

The interaction and observations made during the visit significantly contributed to the students' understanding of real-world caregiving challenges and professional healthcare requirements. The visit further supported the academic and practical foundation of their proposed research study.

The management and staff of the Colombo Caregiver & Health Science School (CCHSS) appreciate the interest shown by the students and wish them success in their academic endeavors.

This letter is issued upon request for academic and official presentation purposes.

Yours faithfully,

Name: G H Chandrika

Designation: Group Assistant Director

Head of Student Affairs,
Accreditations & Business Partnerships
Colombo Caregiver &
Health Science School (Pvt) Ltd

Figure C.1: Visit letters from Remark Elder Care (left) and the Colombo Caregiver & Health Science School (right)

Source: Letters supplied to the group, April 2026.



Figure C.2: Training room at CCHSS (left) and a residents' lounge with faces blurred (right) (Original in Color)

Source: Photographs taken during the group visits, 2026.

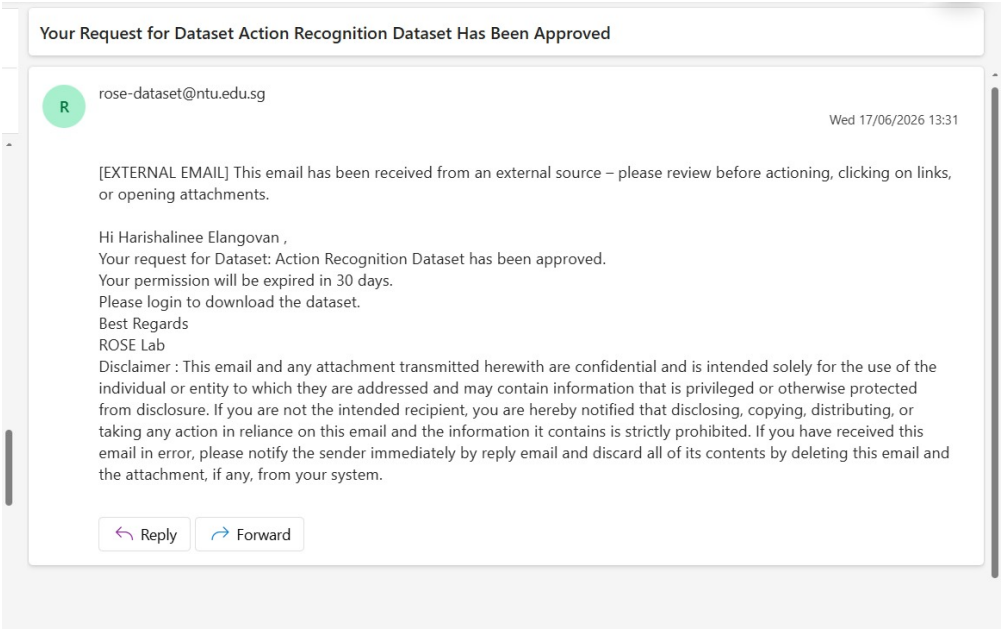
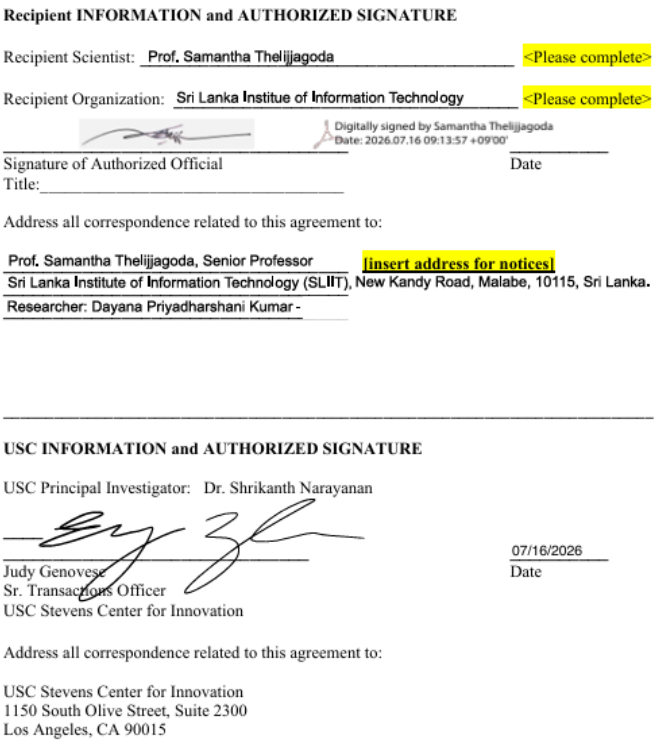


Figure C.3: Approval of access to the NTU RGB+D dataset for the SENTRY component

Source: E-mail from the ROSE Lab, Nanyang Technological University, 17th June. 2026.



Recipient INFORMATION and AUTHORIZED SIGNATURE

Recipient Scientist: Prof. Samantha Thelijjagoda <Please complete>

Recipient Organization: Sri Lanka Institue of Information Technology <Please complete>

Digitally signed by Samantha Thelijjagoda
Date: 2026.07.16 09:13:57 +09'00'

Signature of Authorized Official — Date

Title:

Address all correspondence related to this agreement to:

Prof. Samantha Thelijjagoda, Senior Professor [insert address for notices]
Sri Lanka Institute of Information Technology (SLIIT), New Kandy Road, Malabe, 10115, Sri Lanka.
Researcher: Dayana Priyadharshani Kumar -

USC INFORMATION and AUTHORIZED SIGNATURE

USC Principal Investigator: Dr. Shrikanth Narayanan

Judy Genovese — 07/16/2026
Sr. Transactions Officer — Date
USC Stevens Center for Innovation

Address all correspondence related to this agreement to:

USC Stevens Center for Innovation
1150 South Olive Street, Suite 2300
Los Angeles, CA 90015

Figure C.4: Signature page of the TILES-2018 data use agreement used by PULSE

Source: Agreement with the University of Southern California, signed 16th July. 2026.

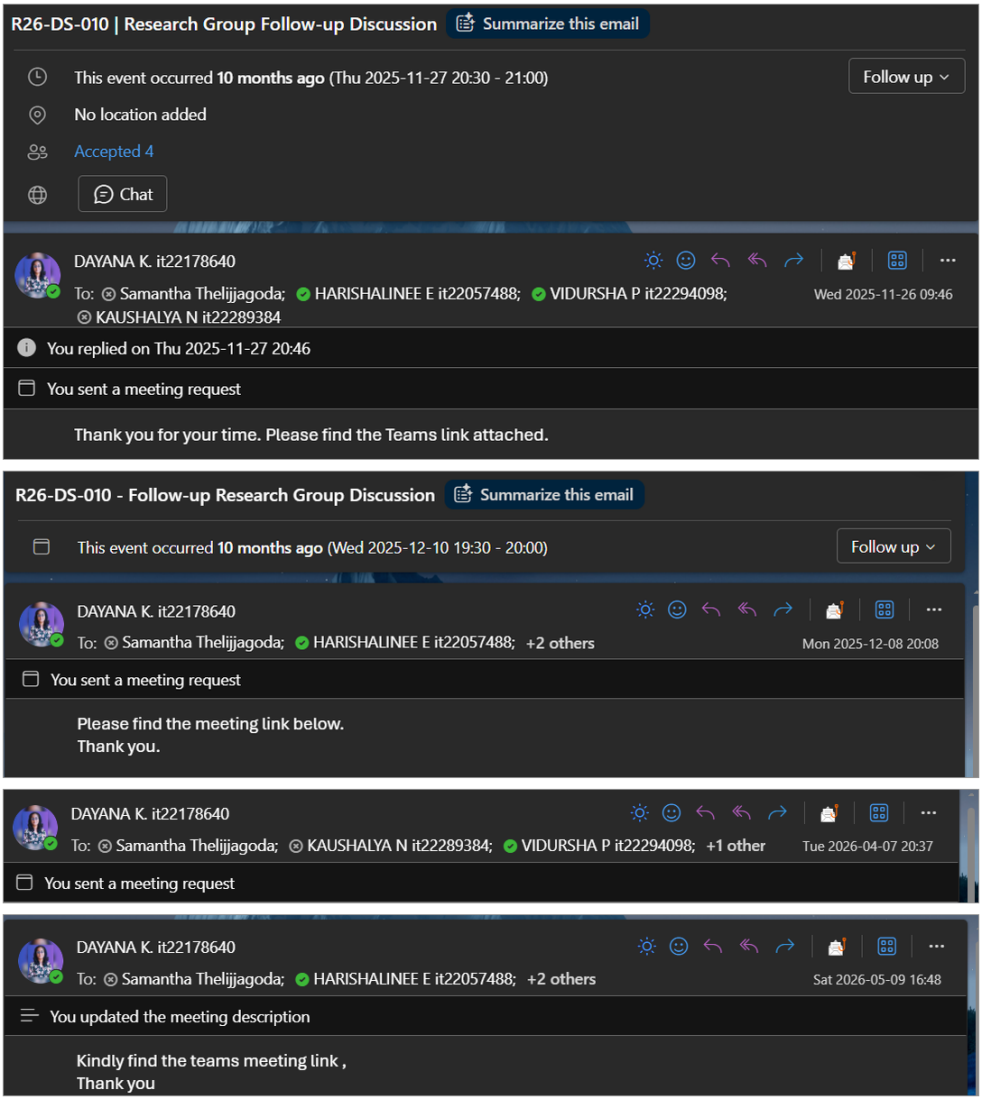


Figure C.5: Teams invitations and mail records, Nov. 2025 to May 2026

Source: Microsoft Teams and Outlook, group account.

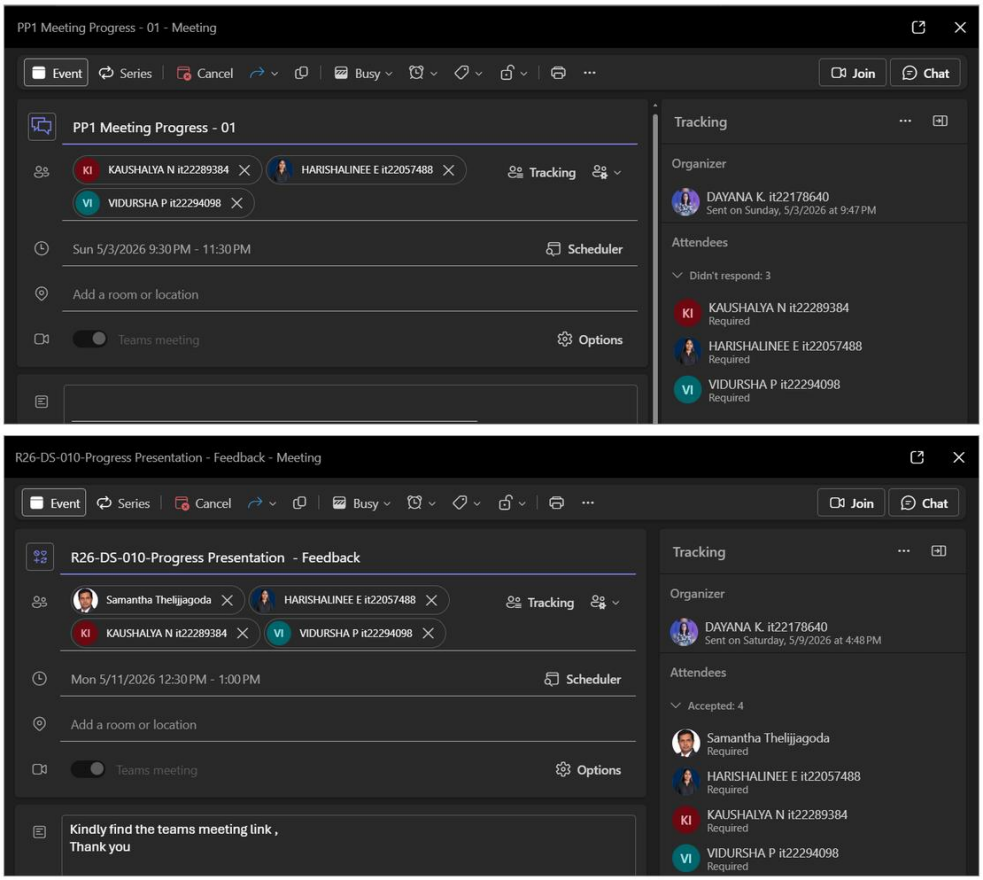


Figure C.6: Group meeting of 3 May 2026 and supervisor feedback meeting of 11 May 2026

Source: Microsoft Teams, group account.

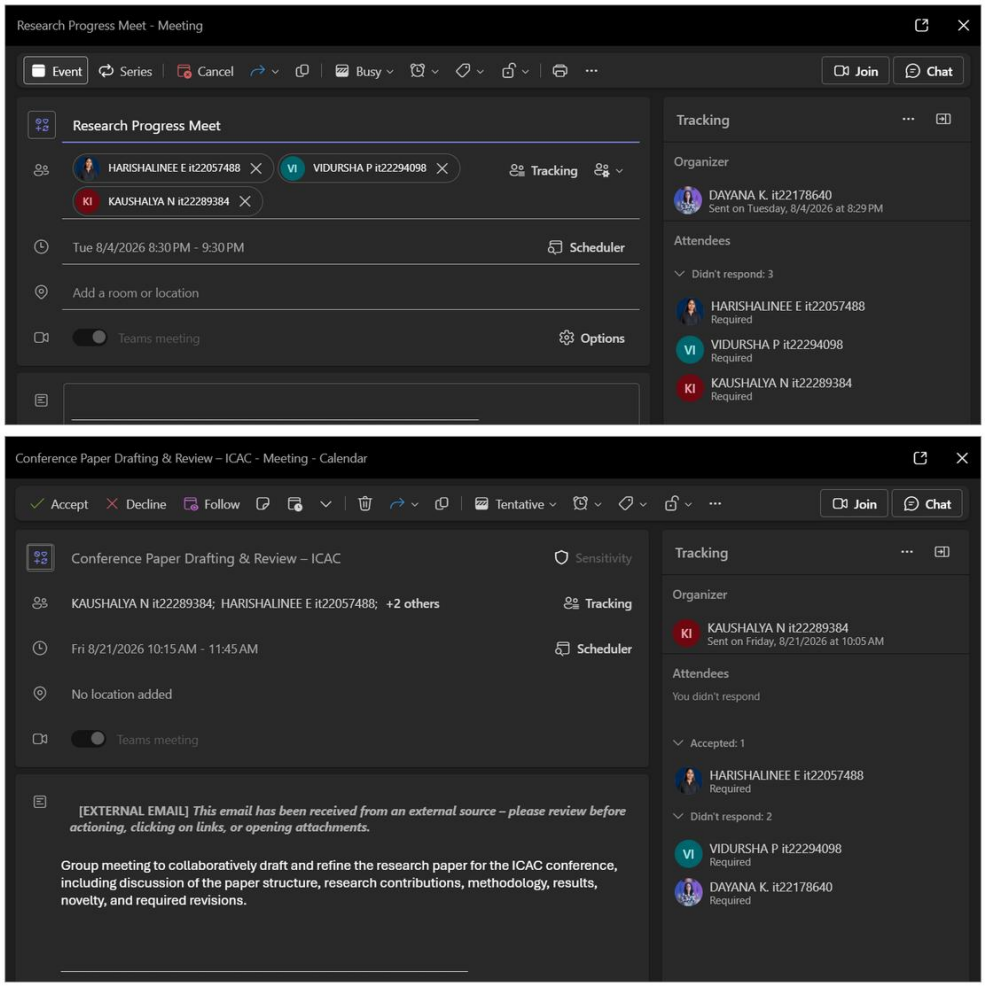


Figure C.7: Research progress meeting of 4 Aug. 2026 and paper drafting session of 21 Aug. 2026

Source: Microsoft Teams, group account.

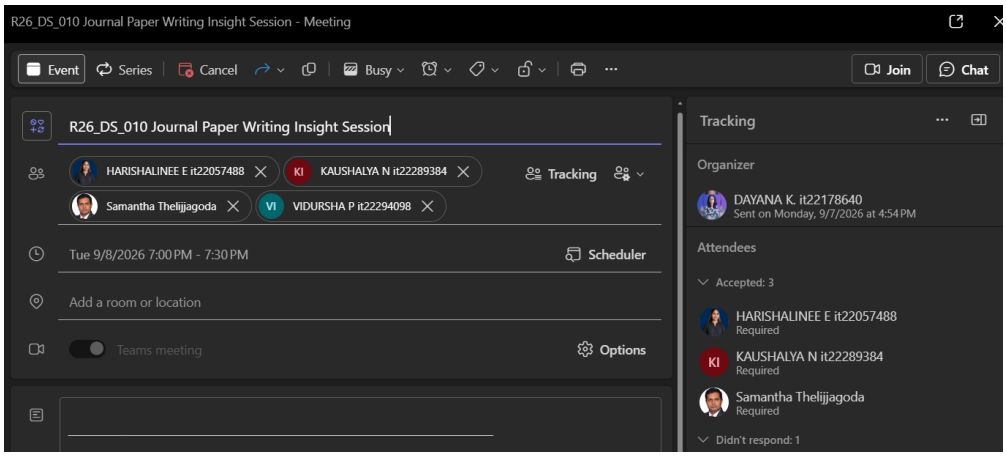


Figure C.8: Journal paper writing session of 8 Sep. 2026

Source: Microsoft Teams, group account.

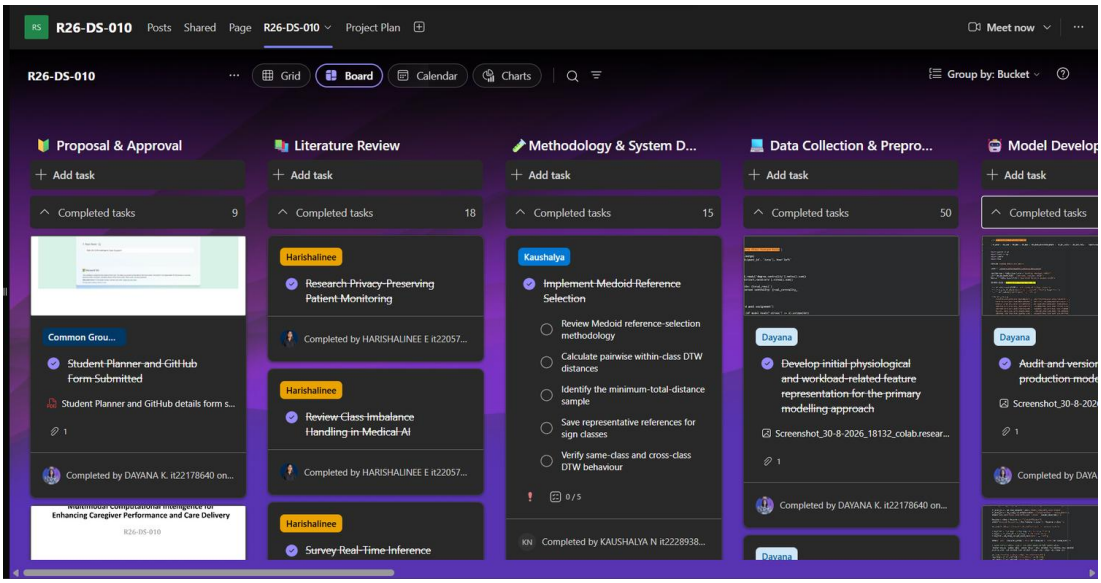


Figure C.9: Planner buckets for proposal, literature review, methodology and data collection

Source: Screenshot of the group's Microsoft Planner board, 2026.

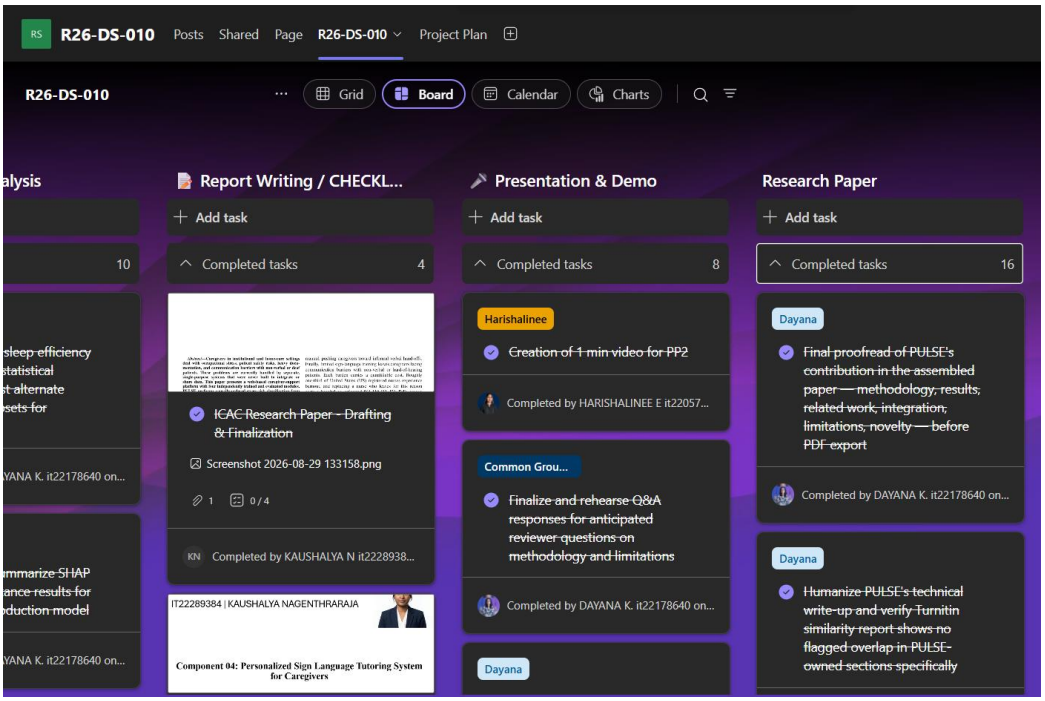


Figure C.10: Planner buckets for report writing, presentations and the research paper

Source: Screenshot of the group's Microsoft Planner board, 2026.

APPENDIX D: REPOSITORY EVIDENCE

Figure D.1 and Figure D.2 show the branch of each component in the project repository, and Figure D.3 shows the integration branch into which they were merged.

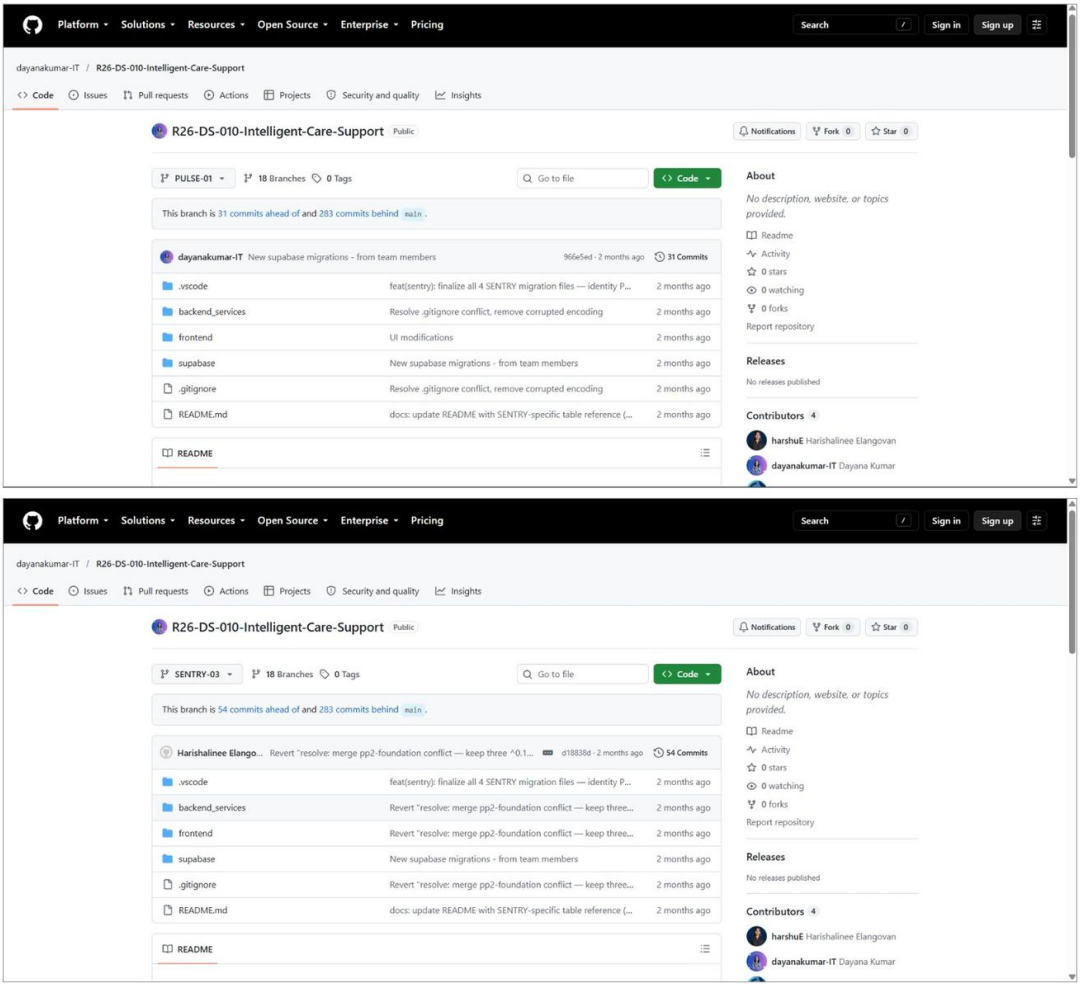


Figure D.1: Repository branches PULSE-01 (above) and SENTRY-03 (below)

Source: Screenshots of the project repository on GitHub, Oct. 2026.

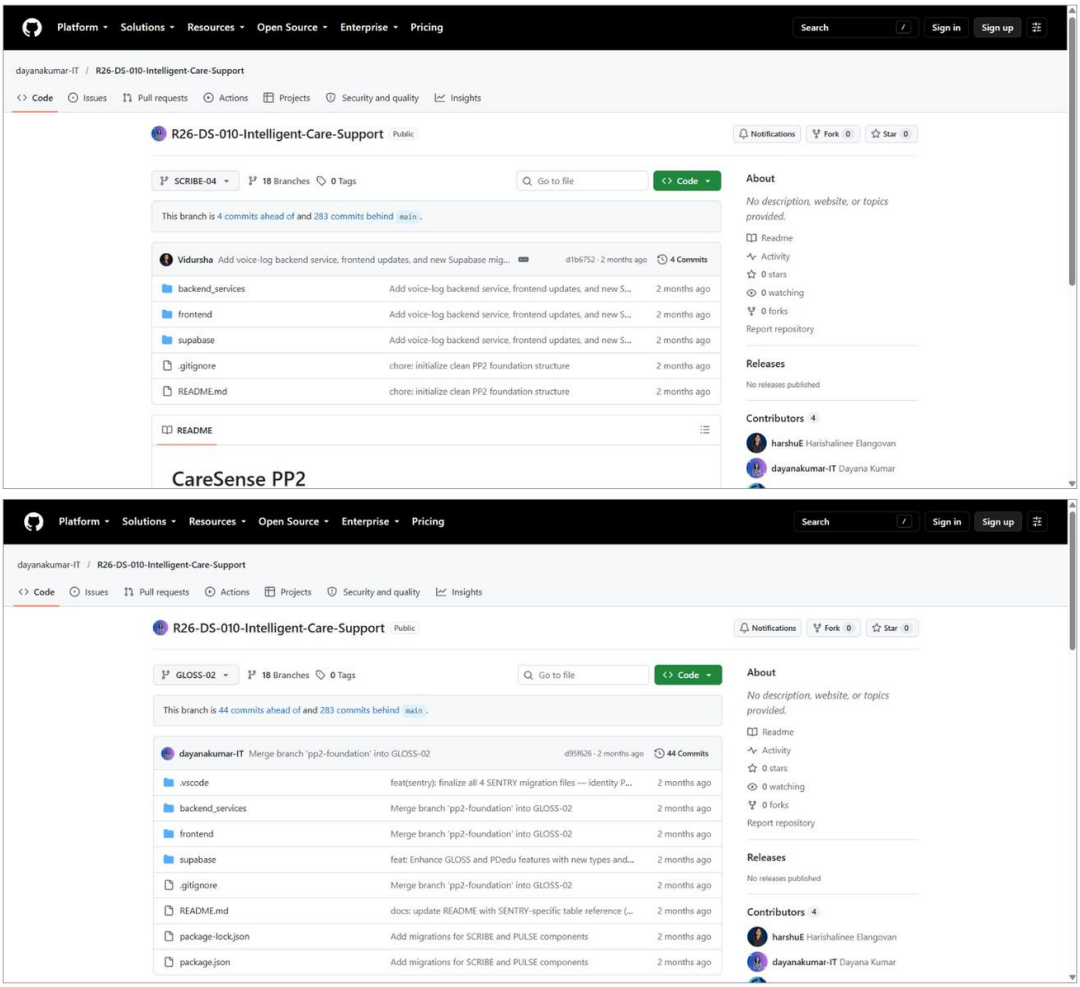


Figure D.2: Repository branches SCRIBE-04 (above) and GLOSS-02 (below)

Source: Screenshots of the project repository on GitHub, Oct. 2026.

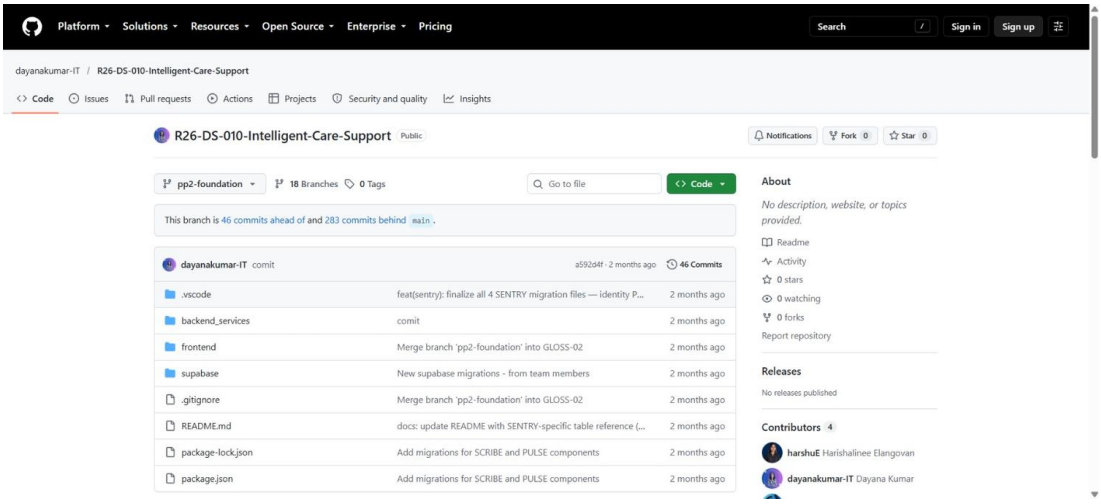


Figure D.3: Integration branch pp2-foundation

Source: Screenshot of the project repository on GitHub, Oct. 2026.

APPENDIX E: ADDITIONAL APPLICATION SCREENS

The screens below complete the picture of the integrated application. The main screens of each module appear in Sections 3.1 and 3.3 to 3.6. All of these were captured from the hosted application in October 2026 using demonstration data, so alert counts and recorded notes are illustrative.

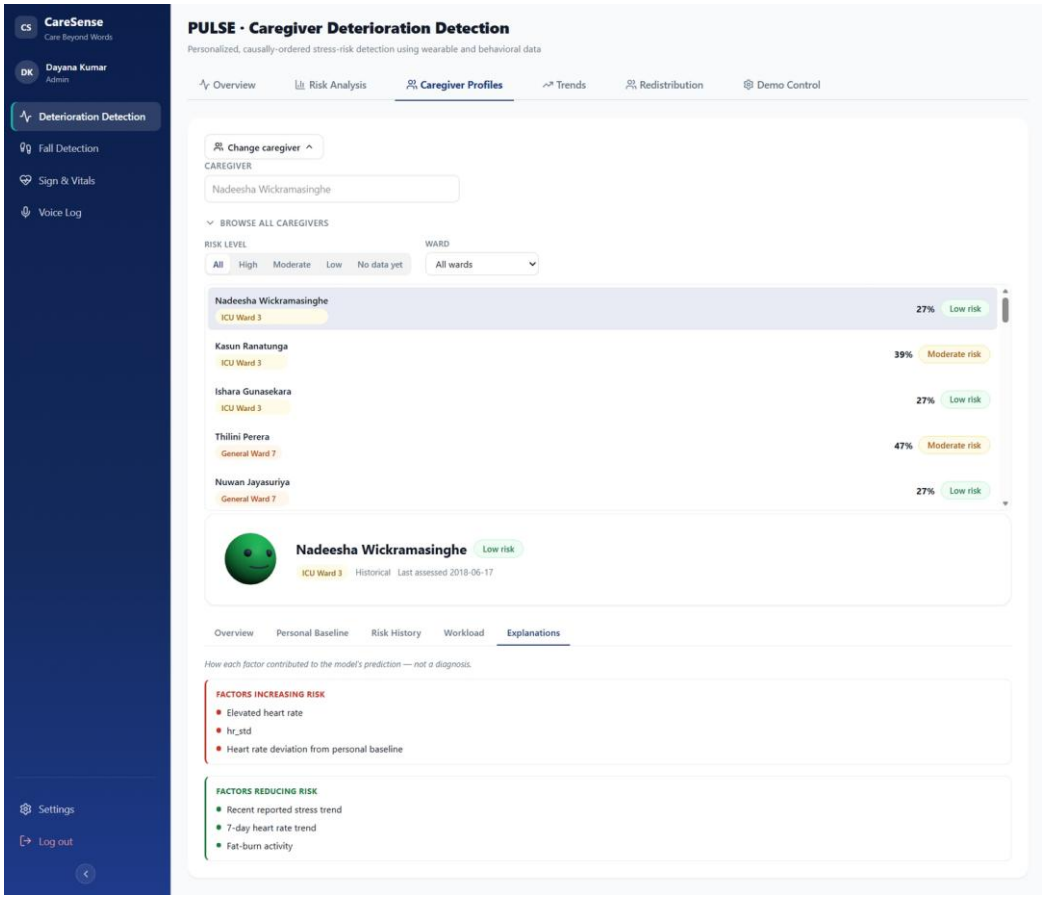


Figure E.1: PULSE caregiver profiles with the explanations tab

Source: Screenshot of the CareSense application, Oct. 2026.



Figure E.2: PULSE team trends, distribution and ward comparison

Source: Screenshot of the CareSense application, Oct. 2026.

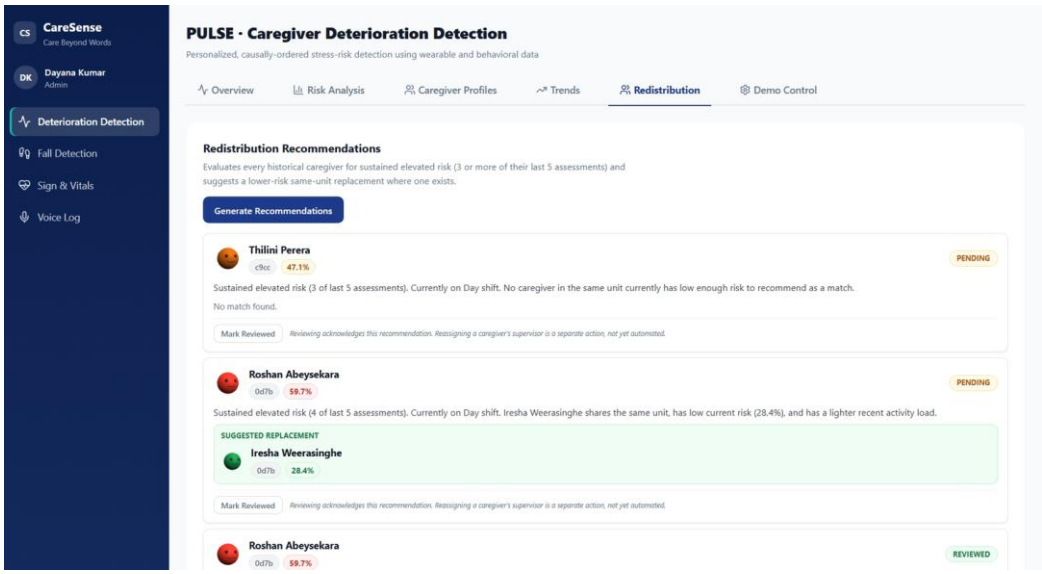


Figure E.3: PULSE redistribution recommendations awaiting review

Source: Screenshot of the CareSense application, Oct. 2026.

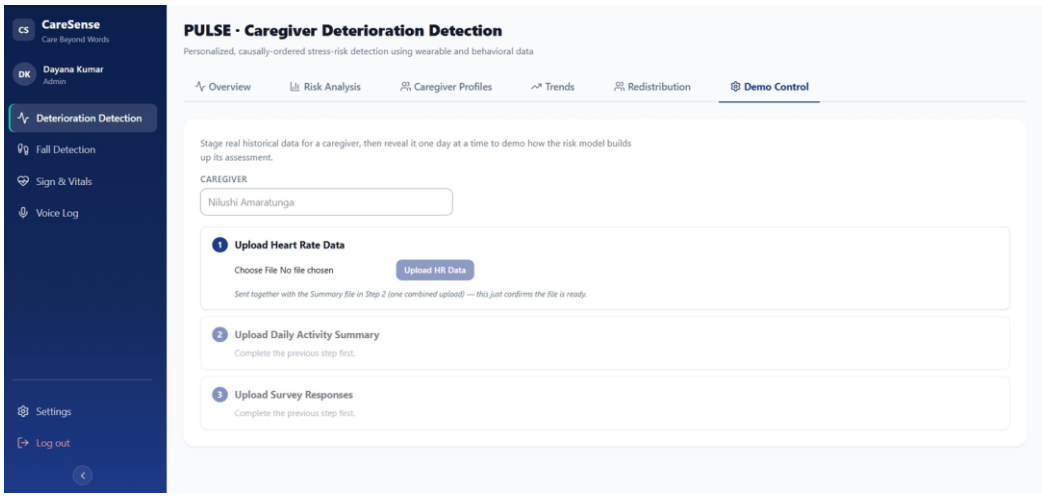


Figure E.4: PULSE demonstration control for staged data uploads

Source: Screenshot of the CareSense application, Oct. 2026.

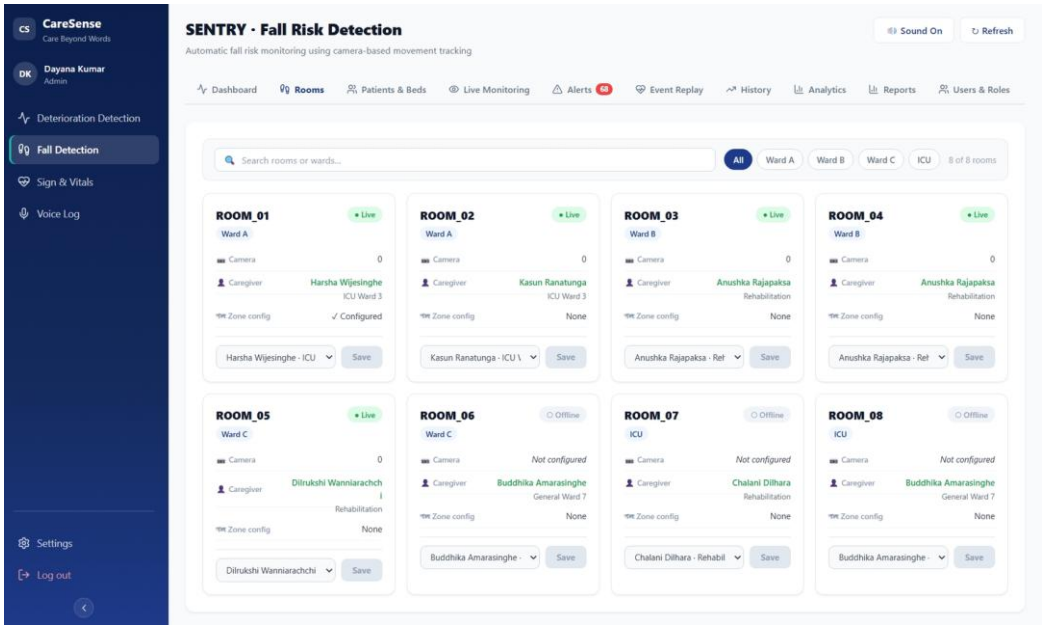


Figure E.5: SENTRY room configuration

Source: Screenshot of the CareSense application, Oct. 2026.

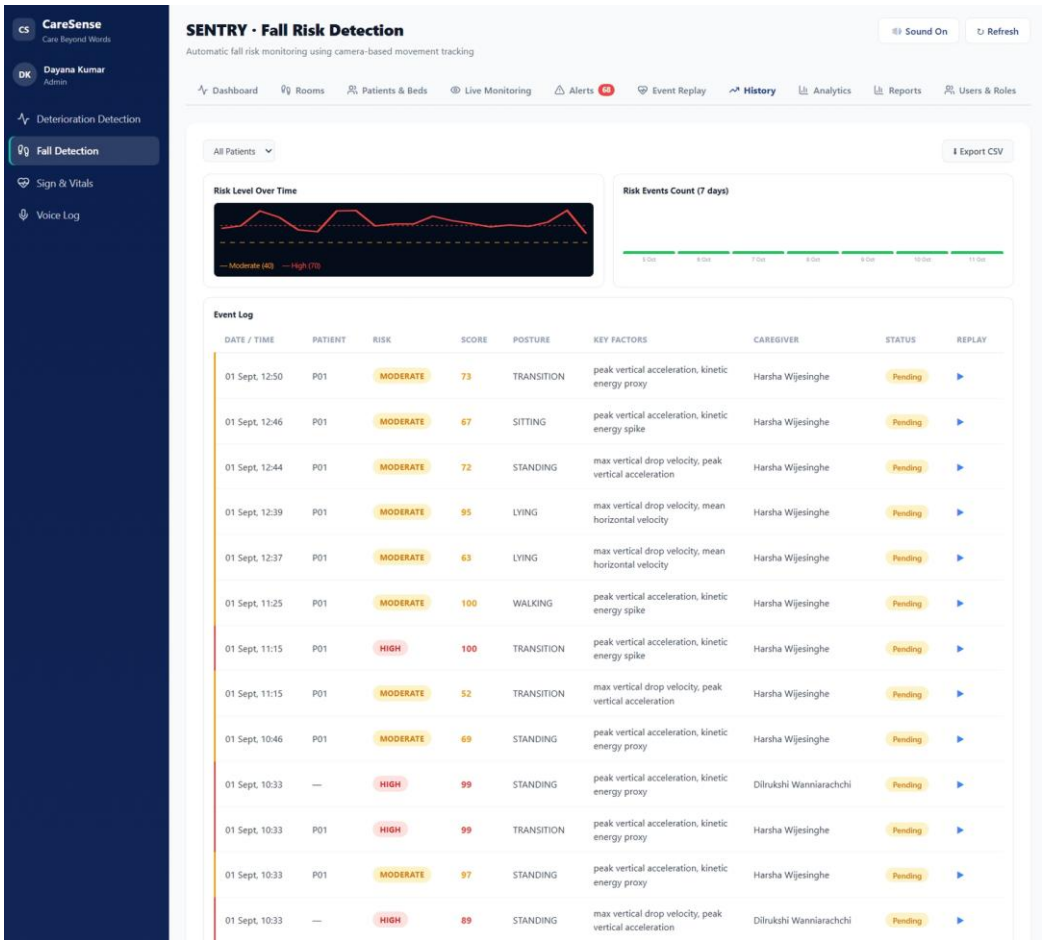


Figure E.6: SENTRY history of risk levels and events

Source: Screenshot of the CareSense application, Oct. 2026.

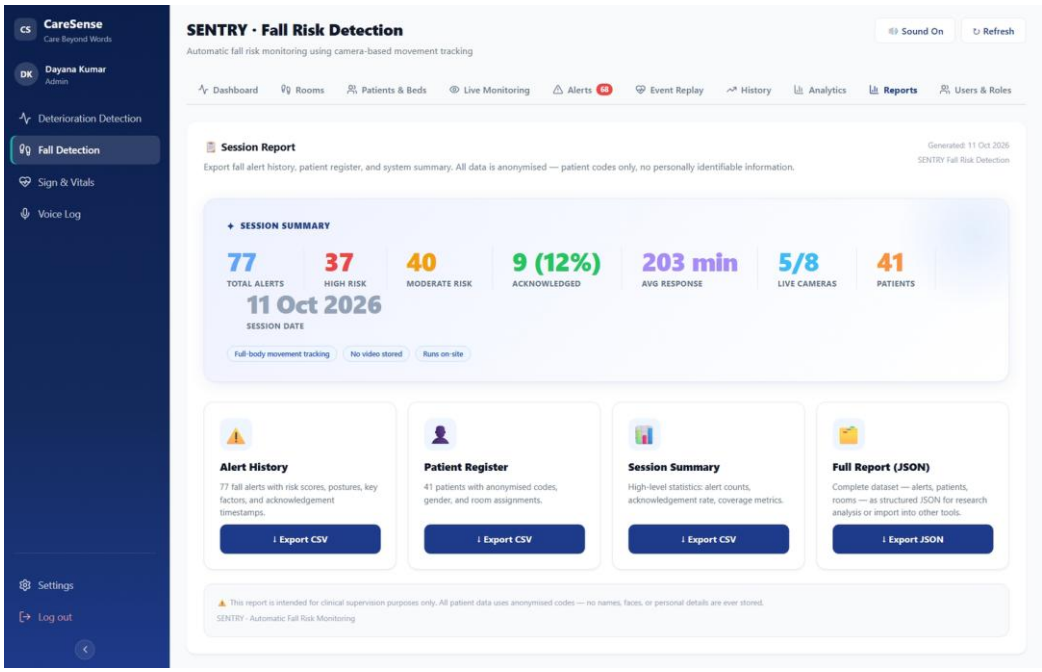


Figure E.7: SENTRY reports with exports and anonymized patient codes

Source: Screenshot of the CareSense application, Oct. 2026.

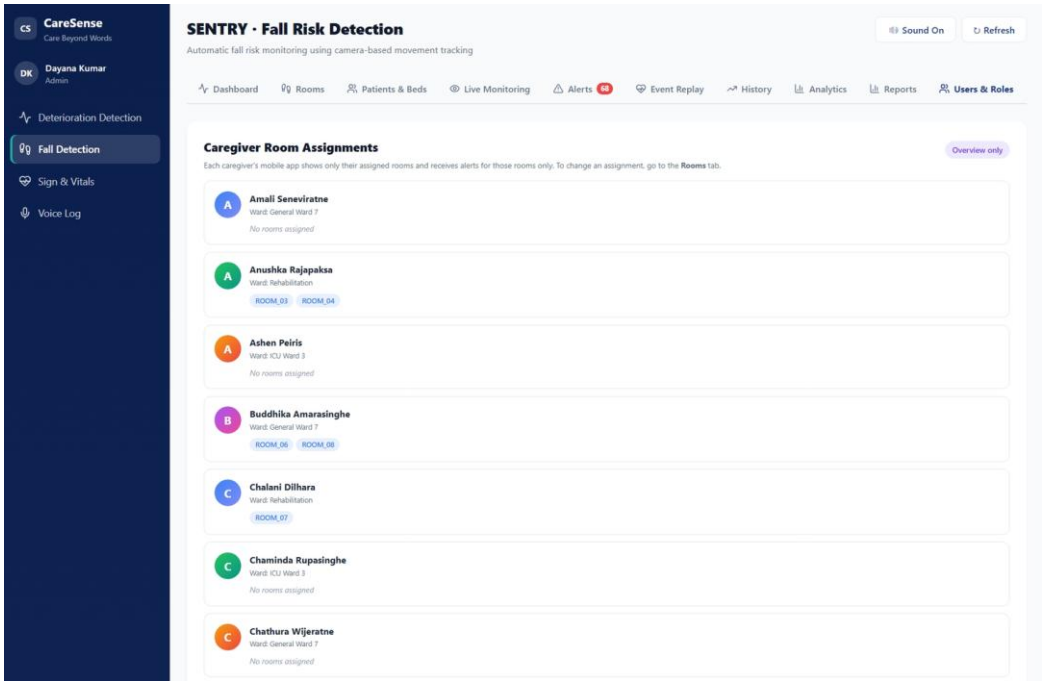


Figure E.8: SENTRY users and roles with caregiver room assignment

Source: Screenshot of the CareSense application, Oct. 2026.

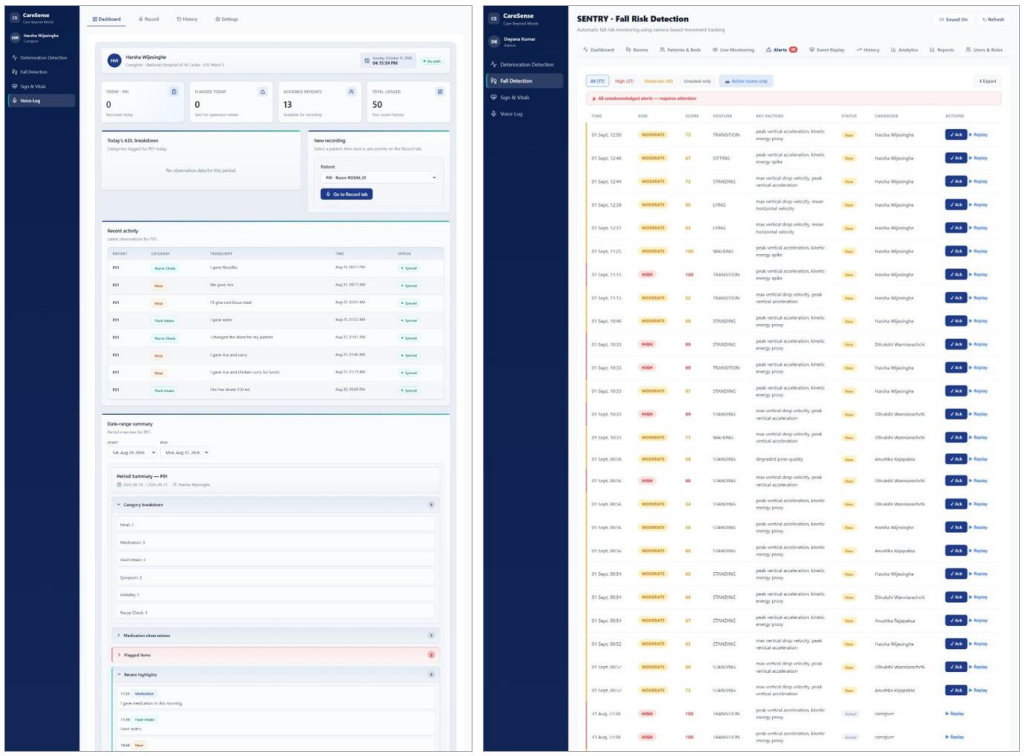


Figure E.9: Mobile views: Voice Log dashboard (left) and SENTRY alerts (right)

Source: Screenshots of the CareSense application, Oct. 2026.

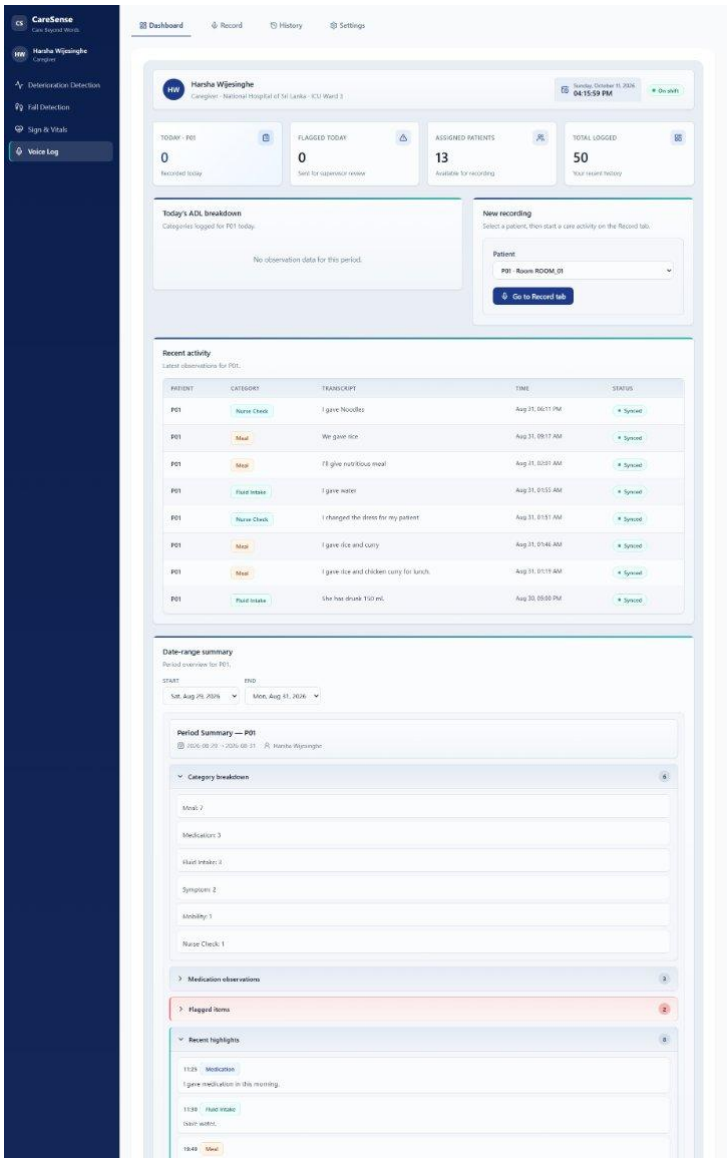


Figure E.10: SCRIBE dashboard summary

Source: Screenshot of the CareSense application, Oct. 2026.



Figure E.11: SCRIBE handover brief

Source: Screenshot of the CareSense application, Oct. 2026.

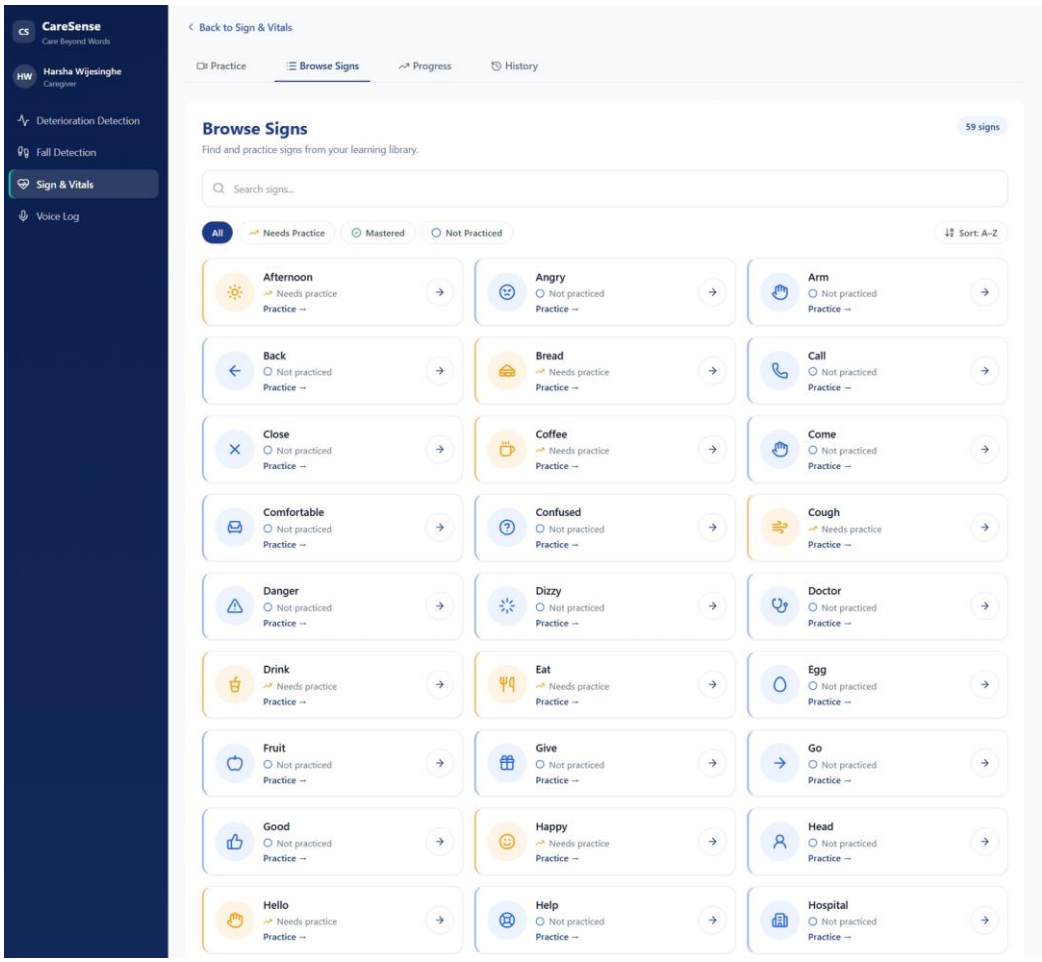


Figure E.12: GLOSS sign library with filters

Source: Screenshot of the CareSense application, Oct. 2026.

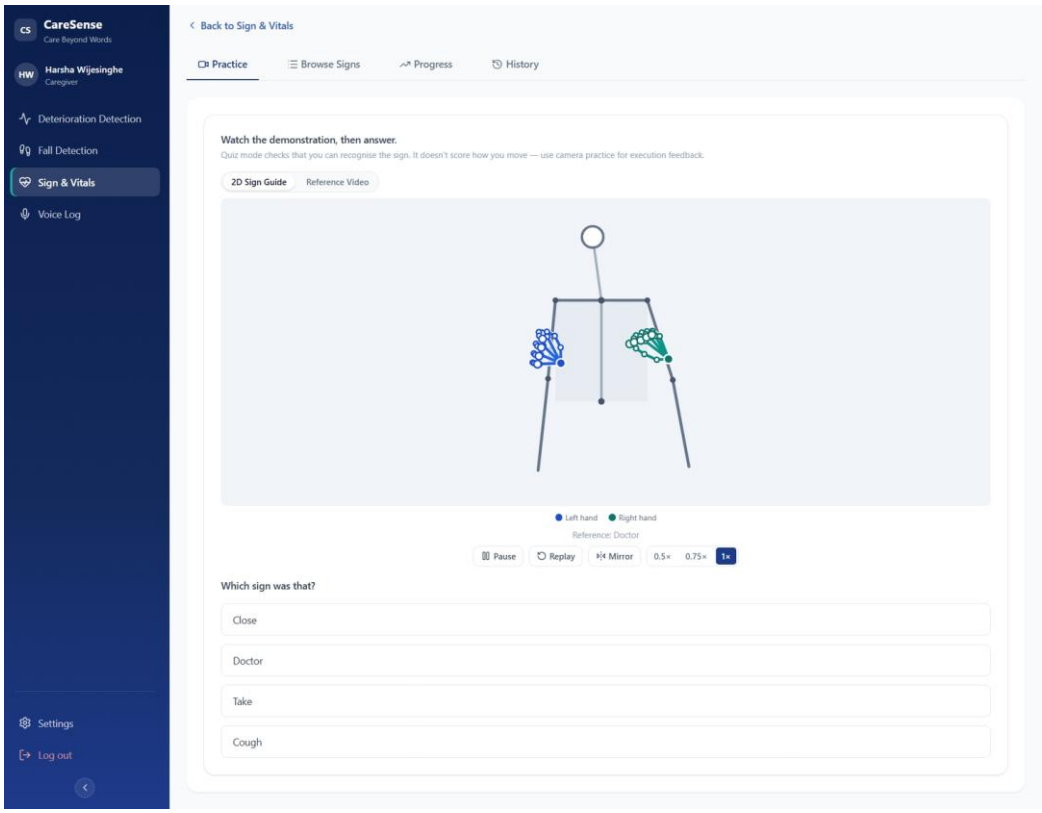


Figure E.13: GLOSS quiz with a 2D sign guide

Source: Screenshot of the CareSense application, Oct. 2026.

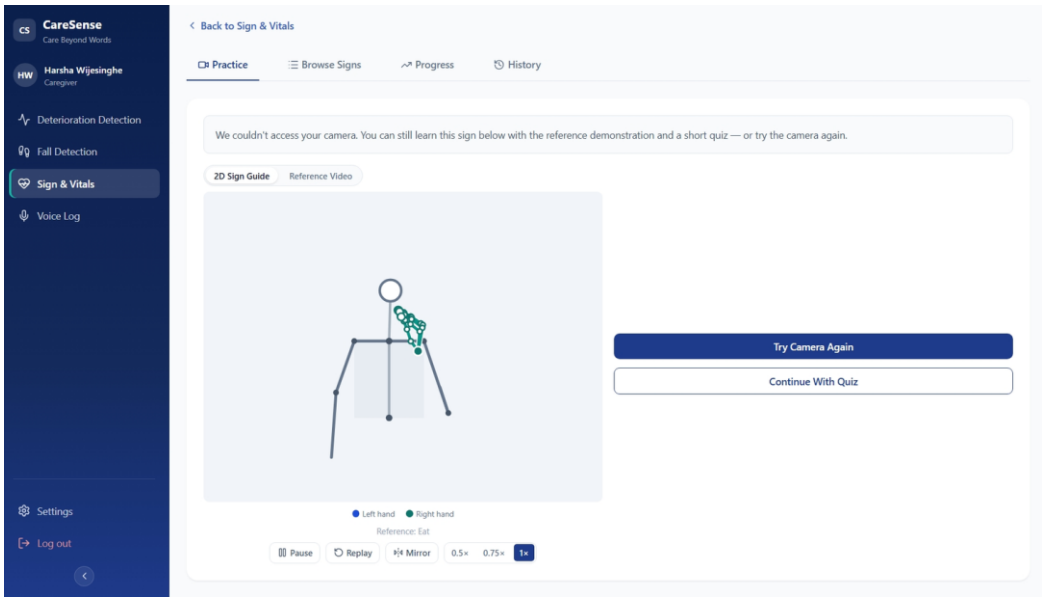


Figure E.14: GLOSS reference demonstration shown when the camera is unavailable

Source: Screenshot of the CareSense application, Oct. 2026.

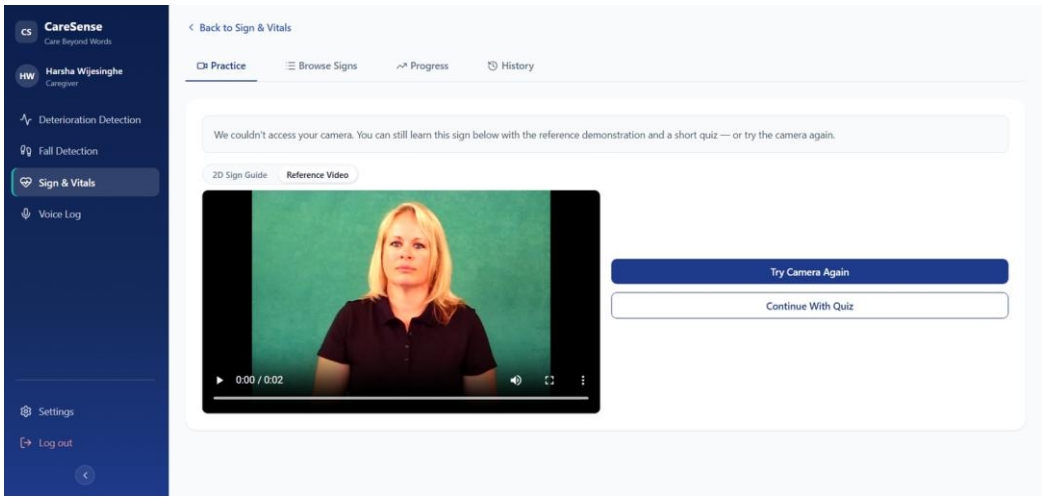


Figure E.15: GLOSS reference video of a sign

Source: Screenshot of the CareSense application, Oct. 2026.

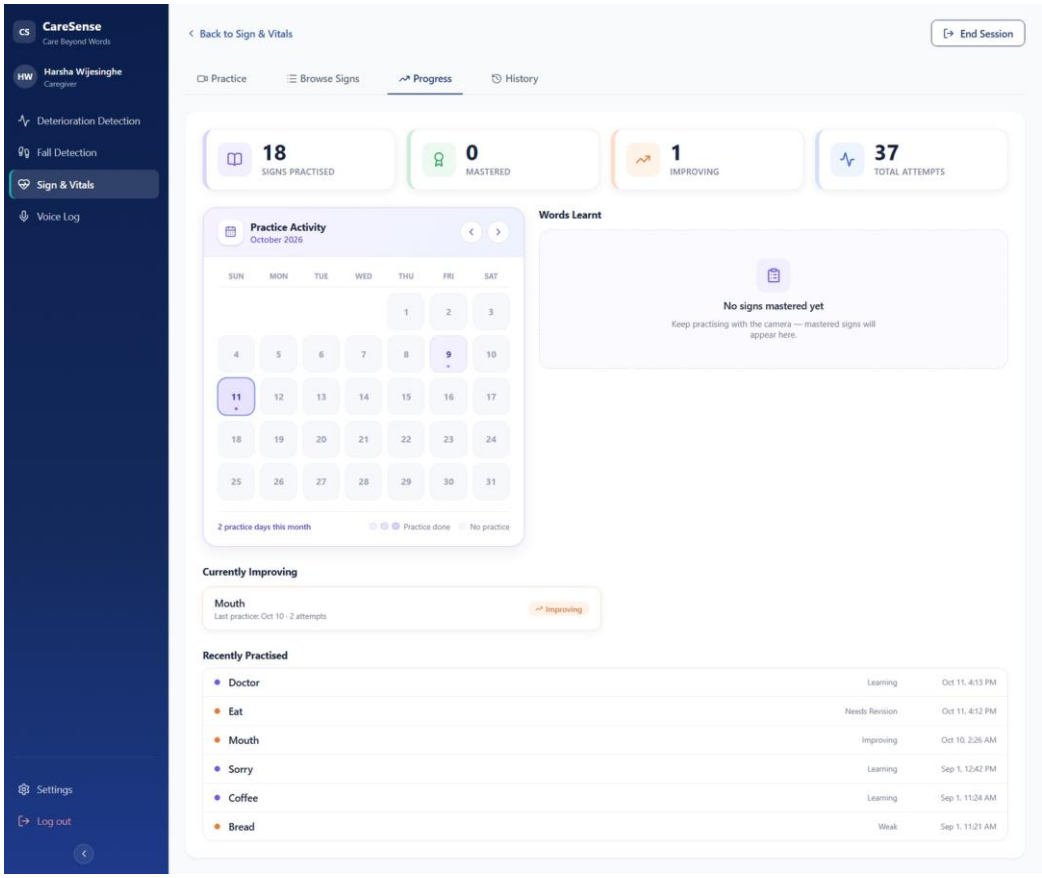


Figure E.16: GLOSS progress and practice calendar

Source: Screenshot of the CareSense application, Oct. 2026.

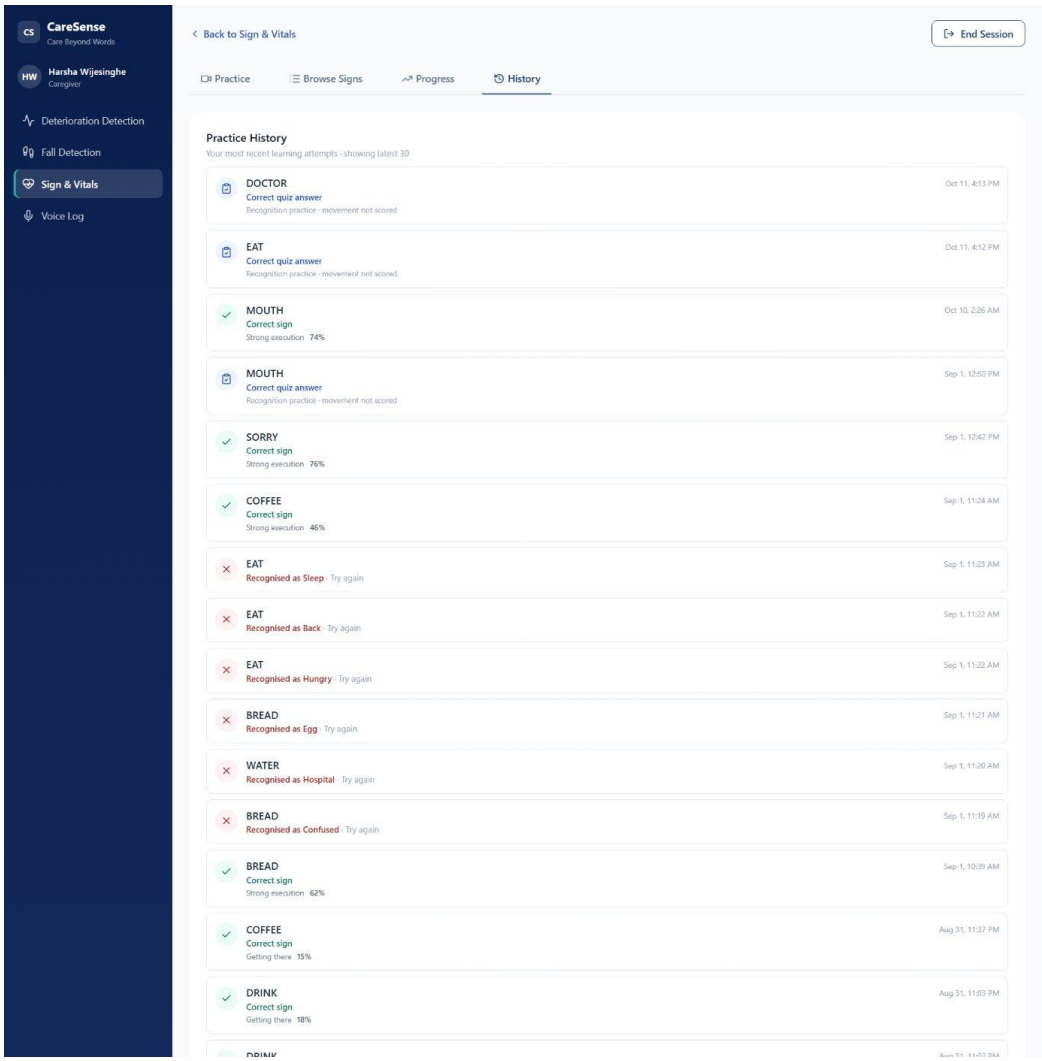


Figure E.17: GLOSS practice history

Source: Screenshot of the CareSense application, Oct. 2026.

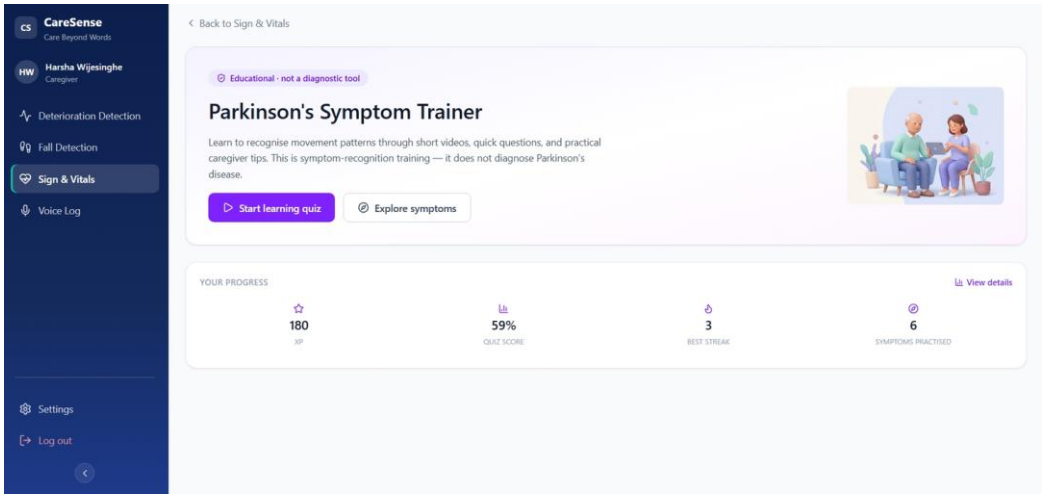


Figure E.18: Parkinson's symptom trainer home (educational, not diagnostic)

Source: Screenshot of the CareSense application, Oct. 2026.

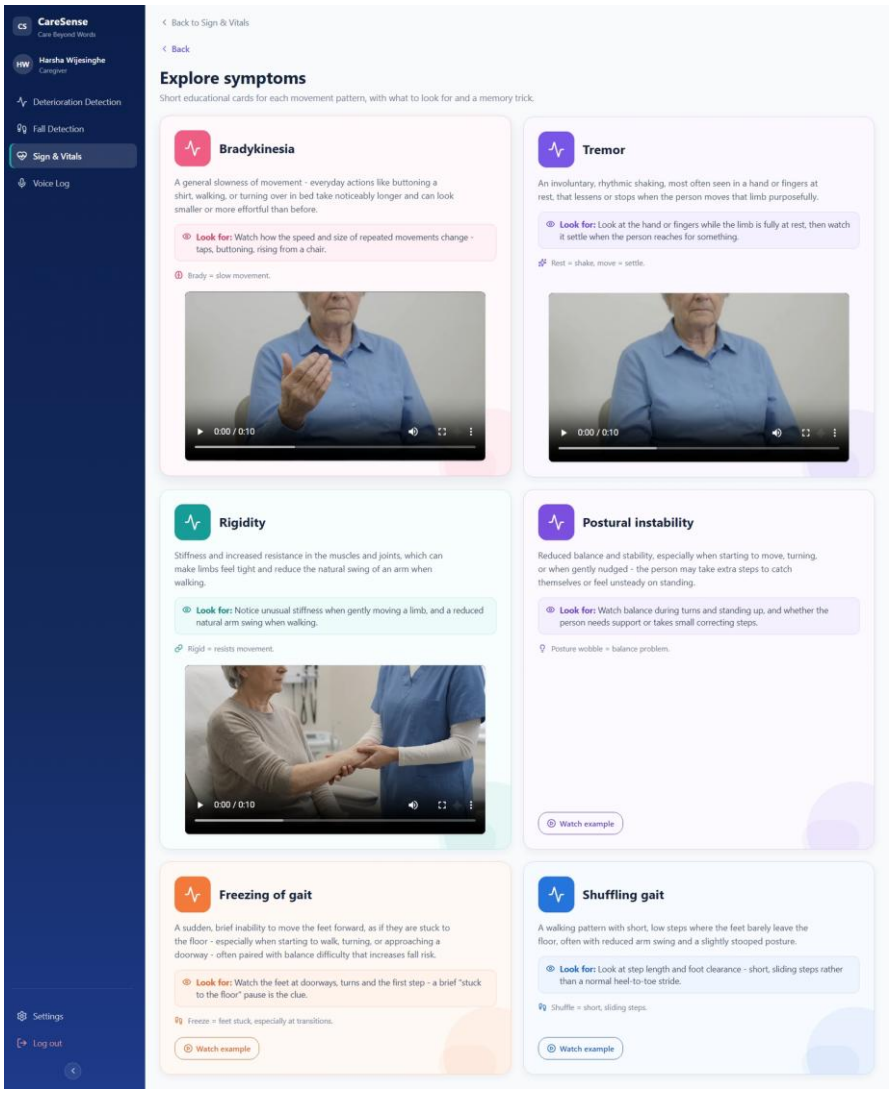


Figure E.19: Parkinson's symptom cards

Source: Screenshot of the CareSense application, Oct. 2026.

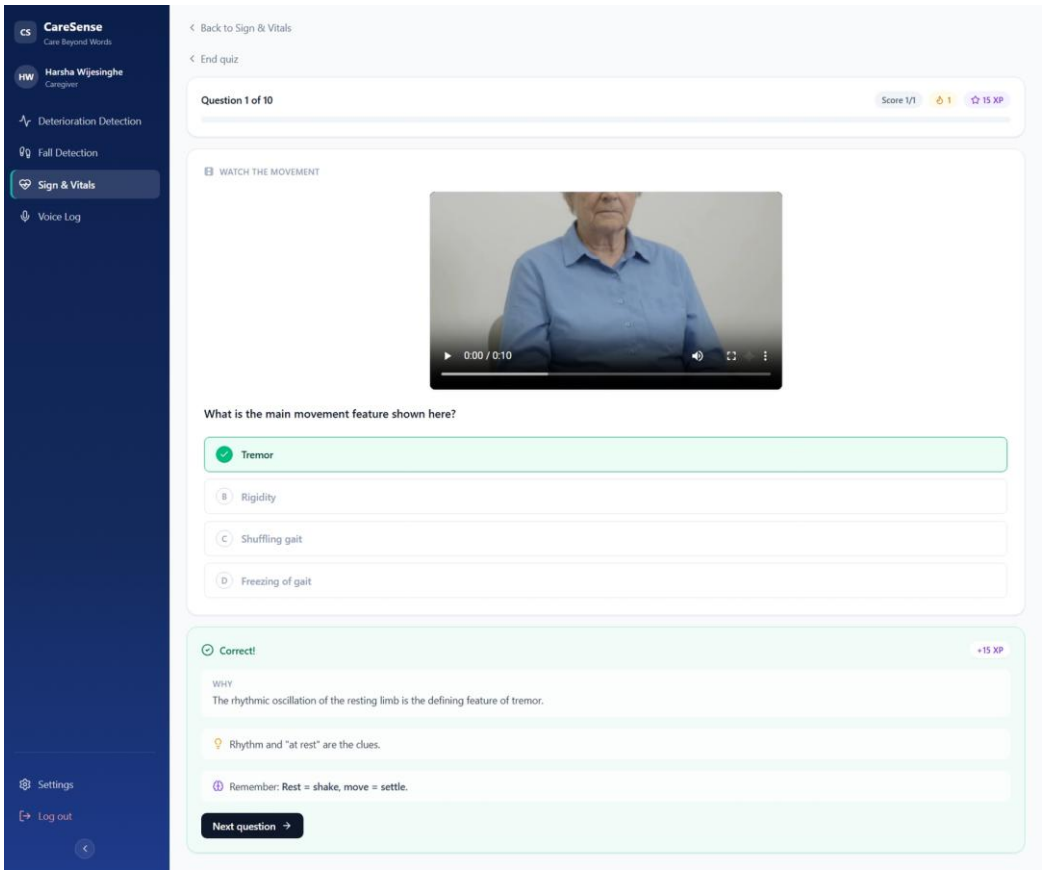


Figure E.20: Parkinson's learning quiz

Source: Screenshot of the CareSense application, Oct. 2026.

APPENDIX F: DISCLOSURE OF AI ASSISTANCE

We hereby declare that:

1. We have read and understood the AI Usage Policy for the Research Project (IT4010).
2. We acknowledge that AI tools are to be used only as support tools, not as substitutes for our own understanding and work.
3. We agree to use AI tools responsibly, ethically, and transparently.
4. We will clearly disclose any use of AI tools in my submissions.
5. We understand that we bear full responsibility and accountability for any content, code, model, or output generated using AI tools.
6. We confirm that we will be able to explain, justify, and defend all aspects of our work during evaluations.
7. We understand that failure to comply with this policy may result in academic penalties.

In line with this declaration, we used AI tools as a support tool during the preparation of this thesis. It helped us to organize the structure against the faculty template, to draft and edit text from our own project notes, to format the document and to check the details of the cited references. The research design, the experiments, the interpretation of the results and the conclusions are our own. We reviewed the text before submission, and we take full responsibility for its content, its claims and its citations.